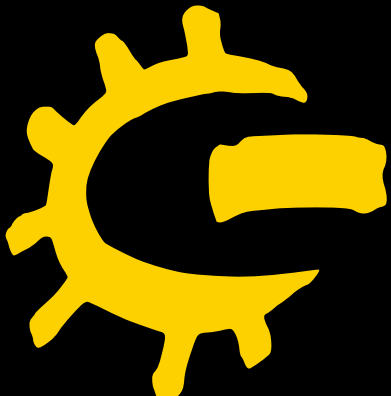
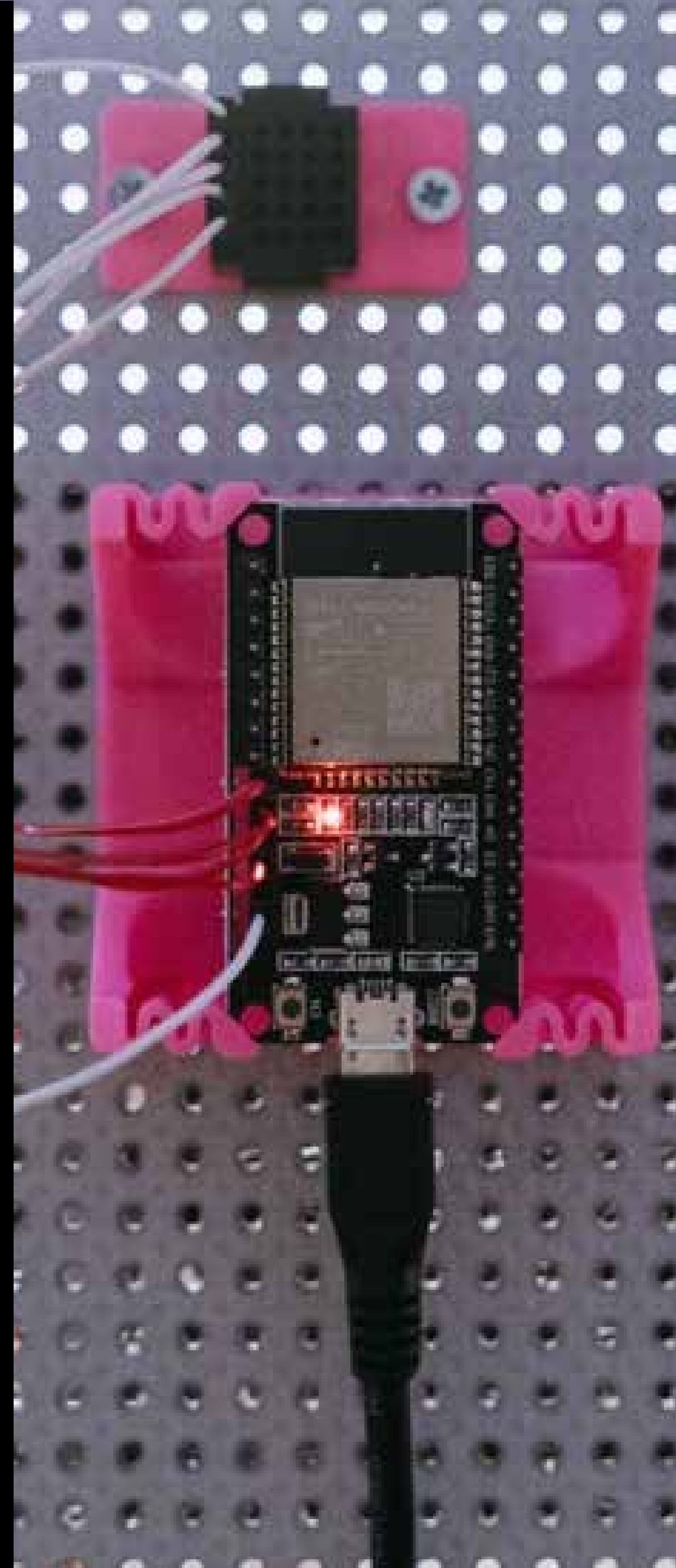


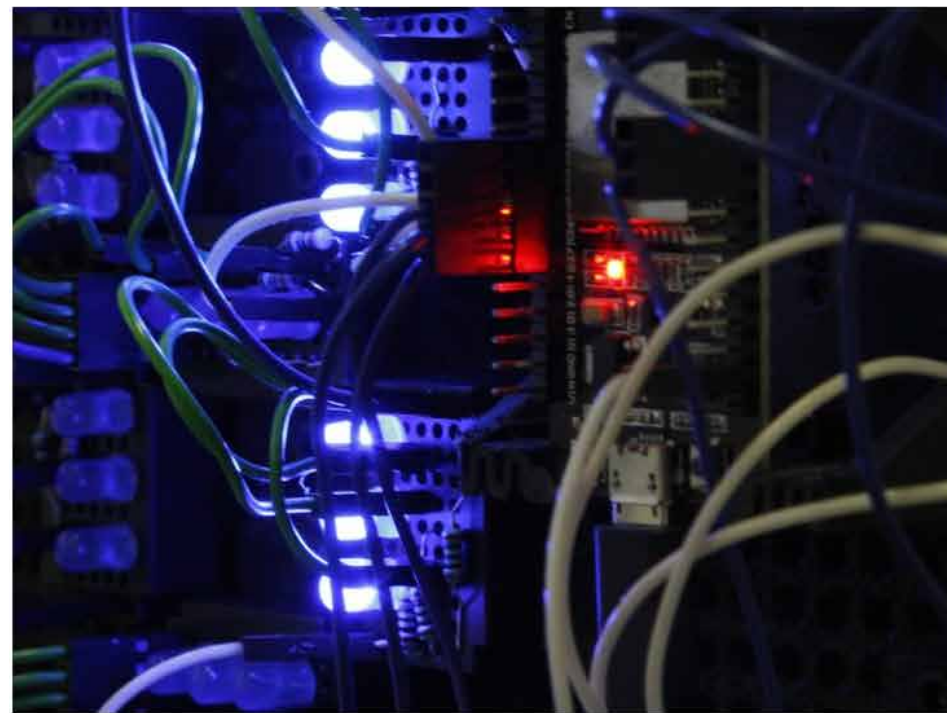
CREATIVE HARDWARE WITH ESP32

Workshop by Silvia Binda Heiseroва

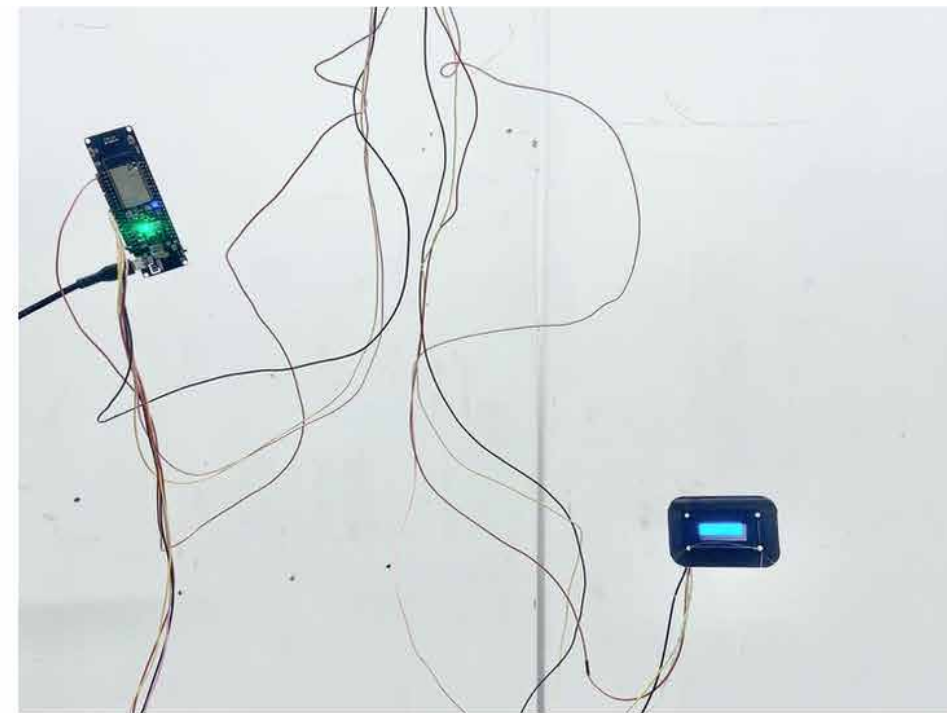
mur  at



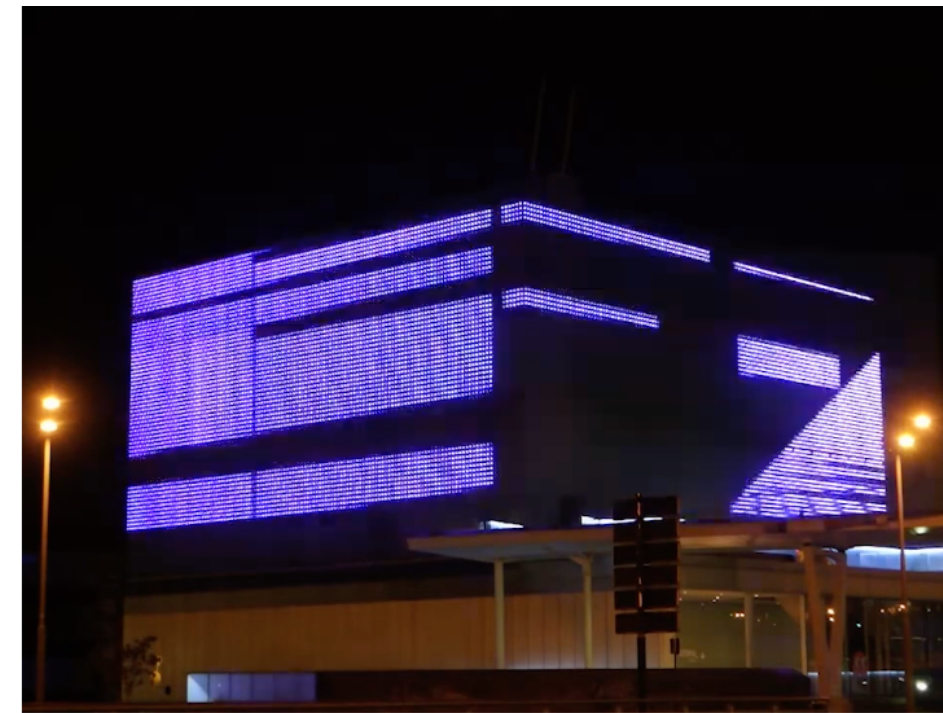
Introduction



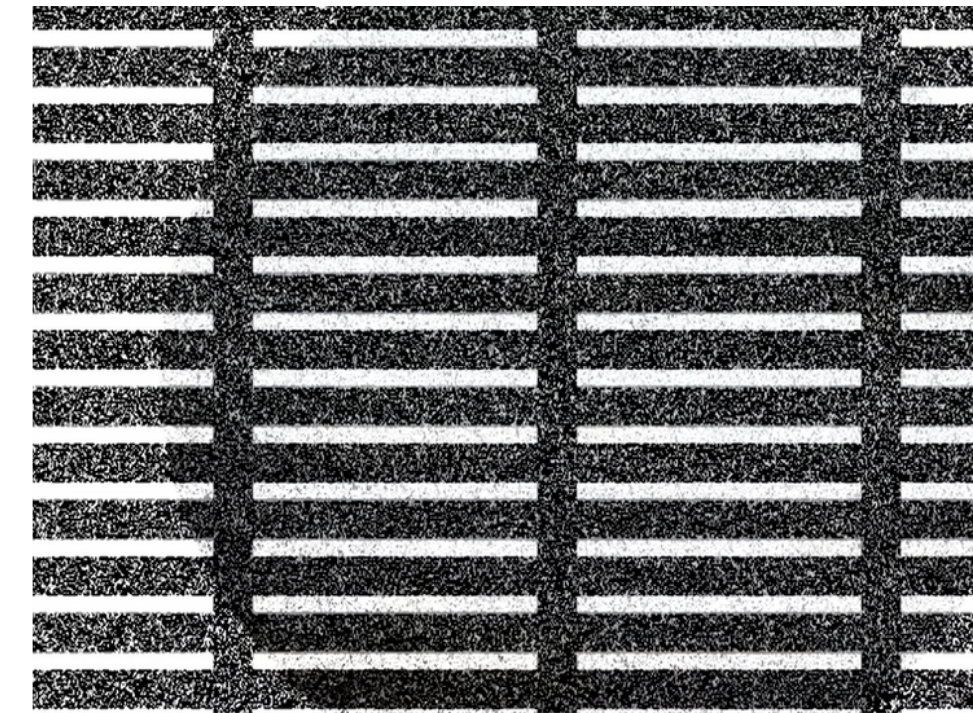
Conversatorio



#Woman Critical Interface



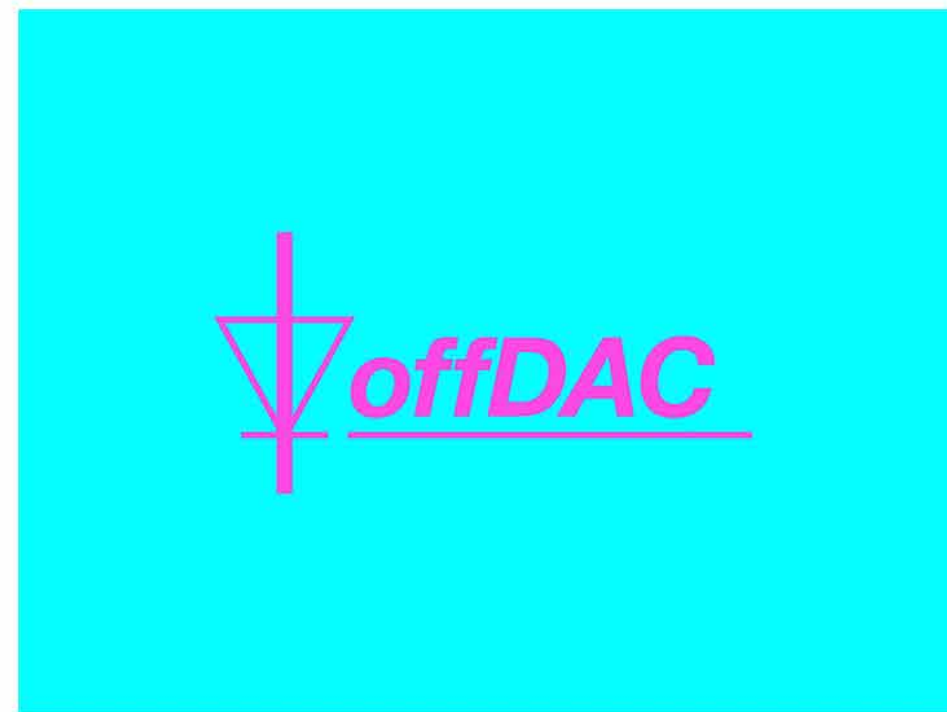
Hyperimage



TEXT0



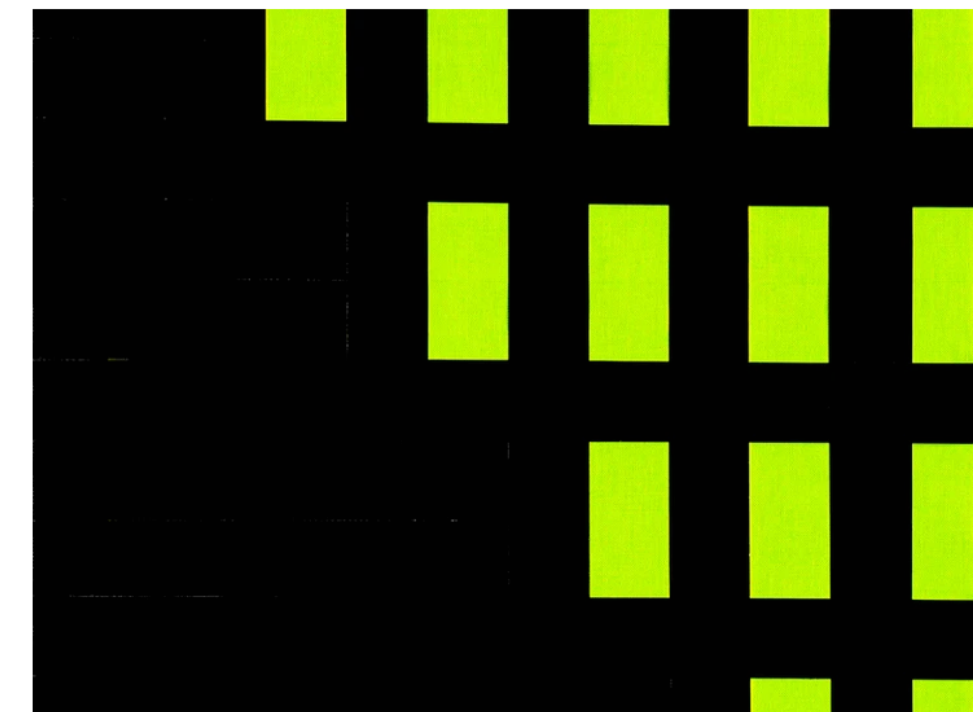
Vending Machine



offDAC

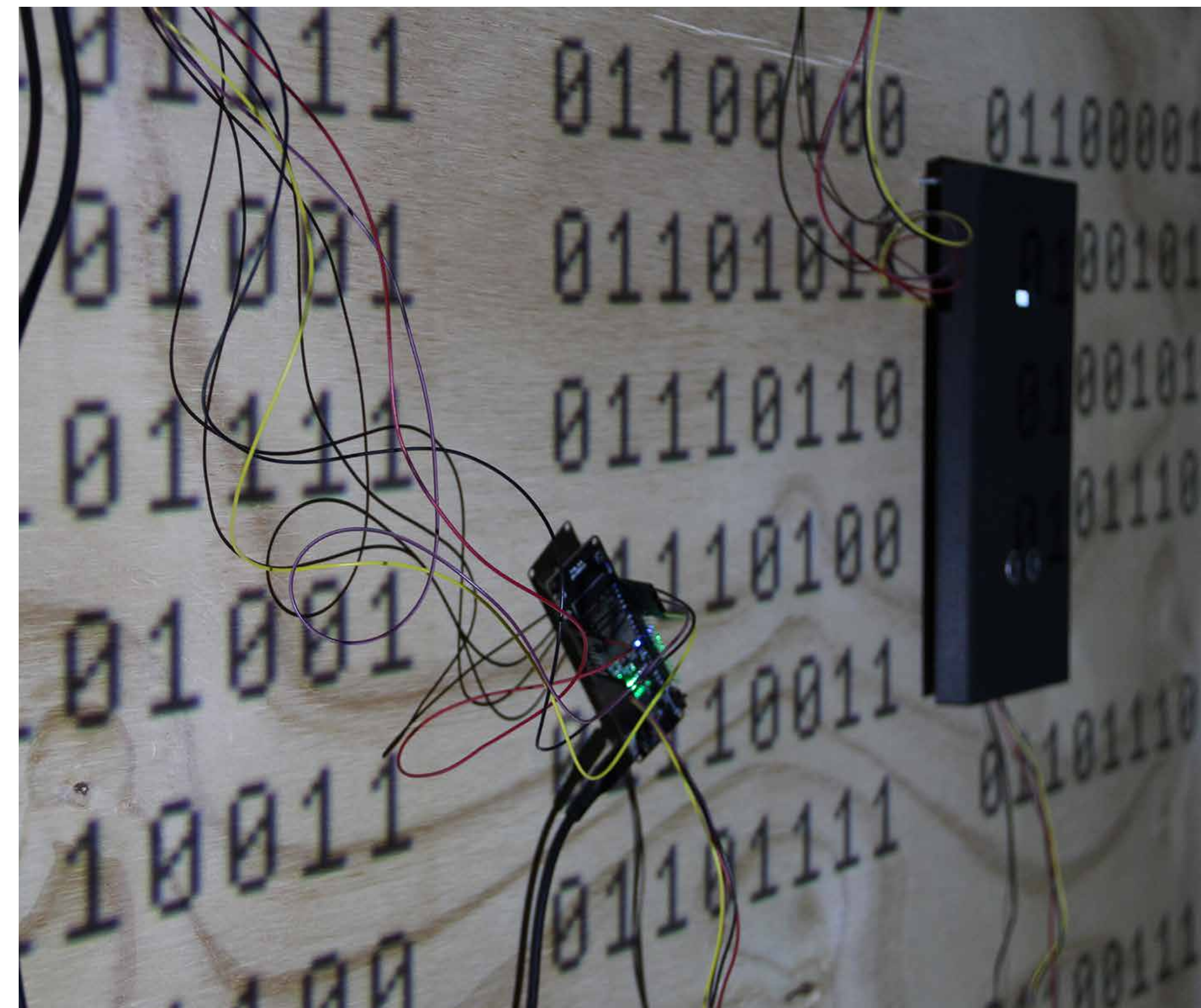
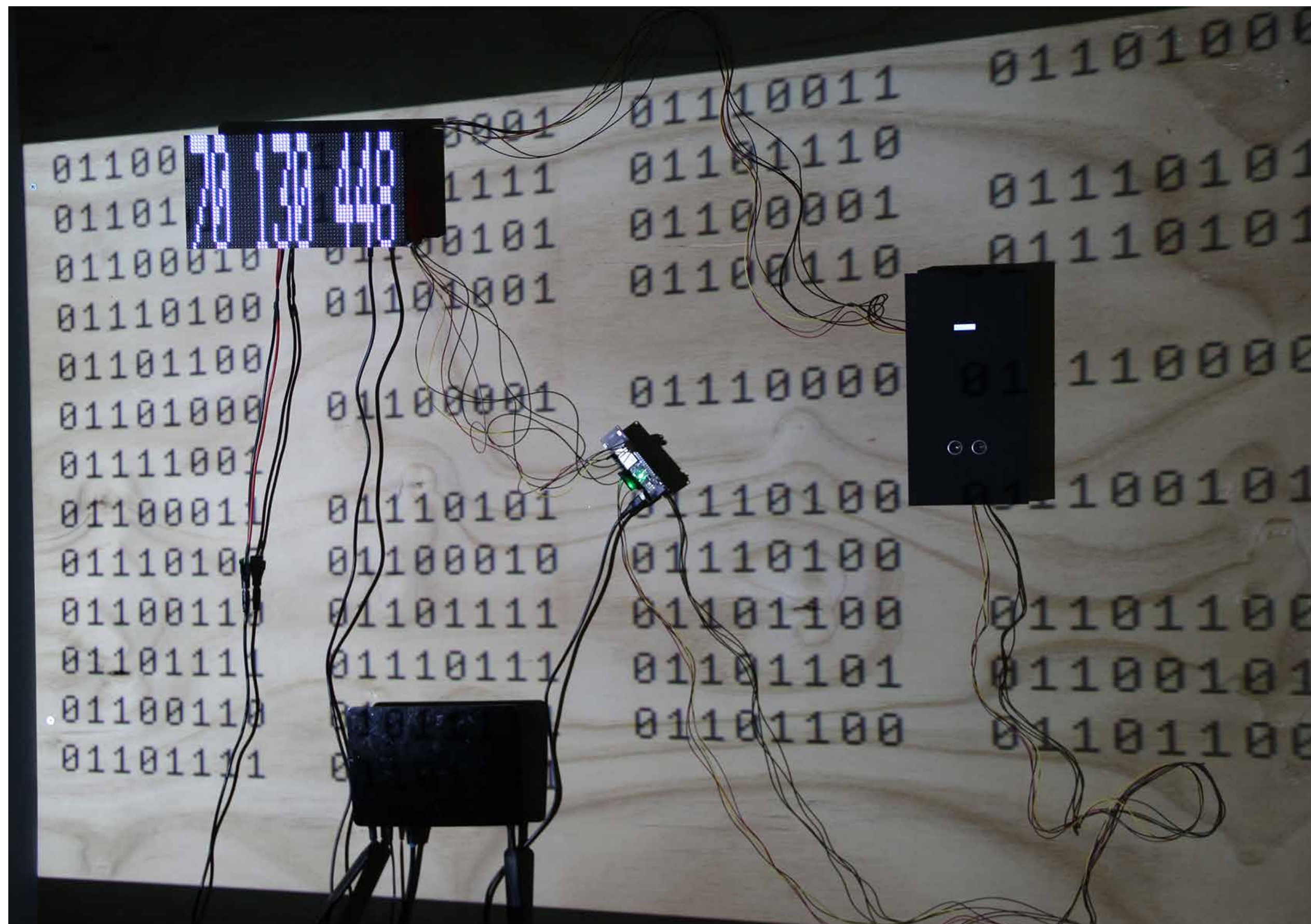


Moving Cityscapes

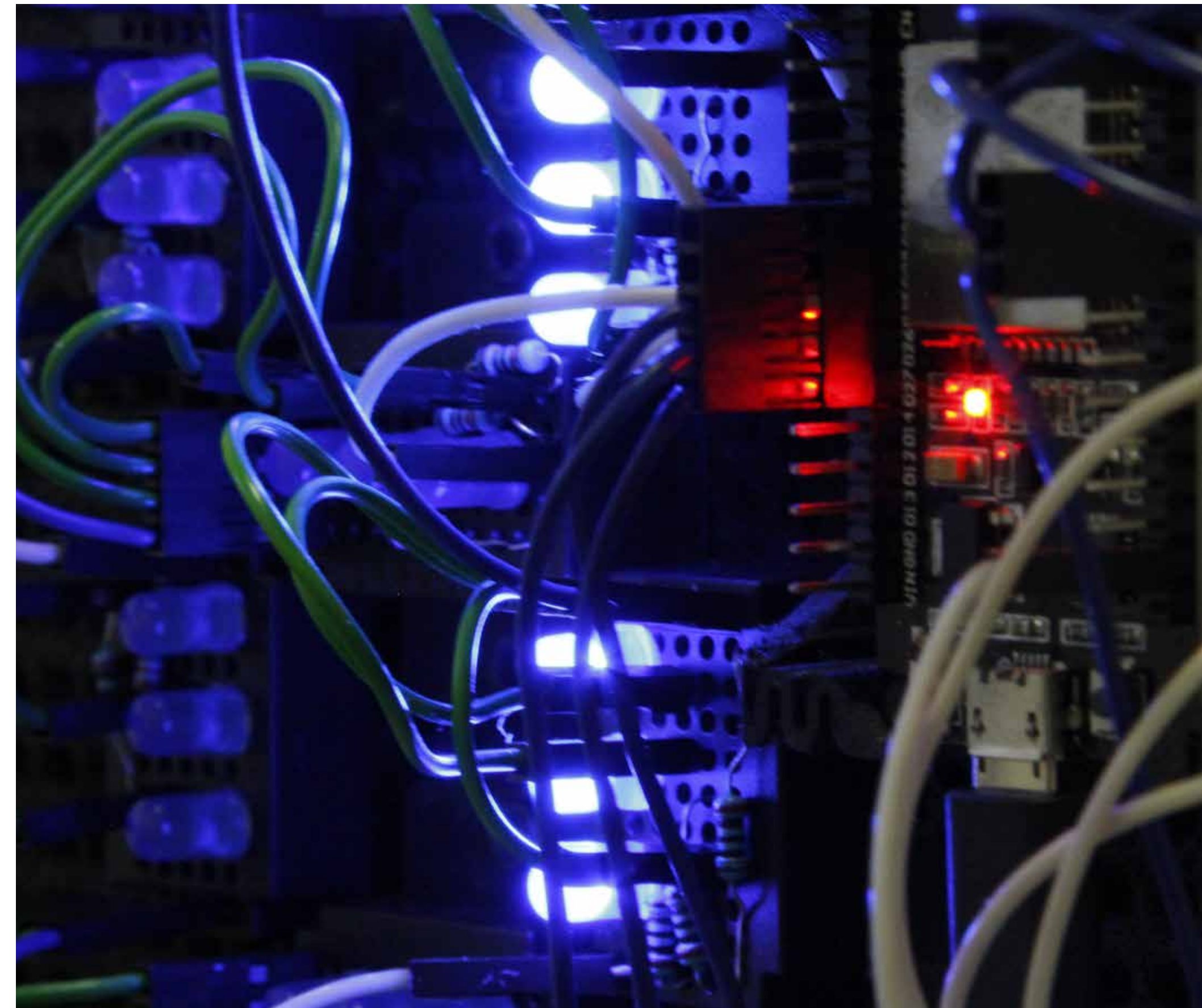
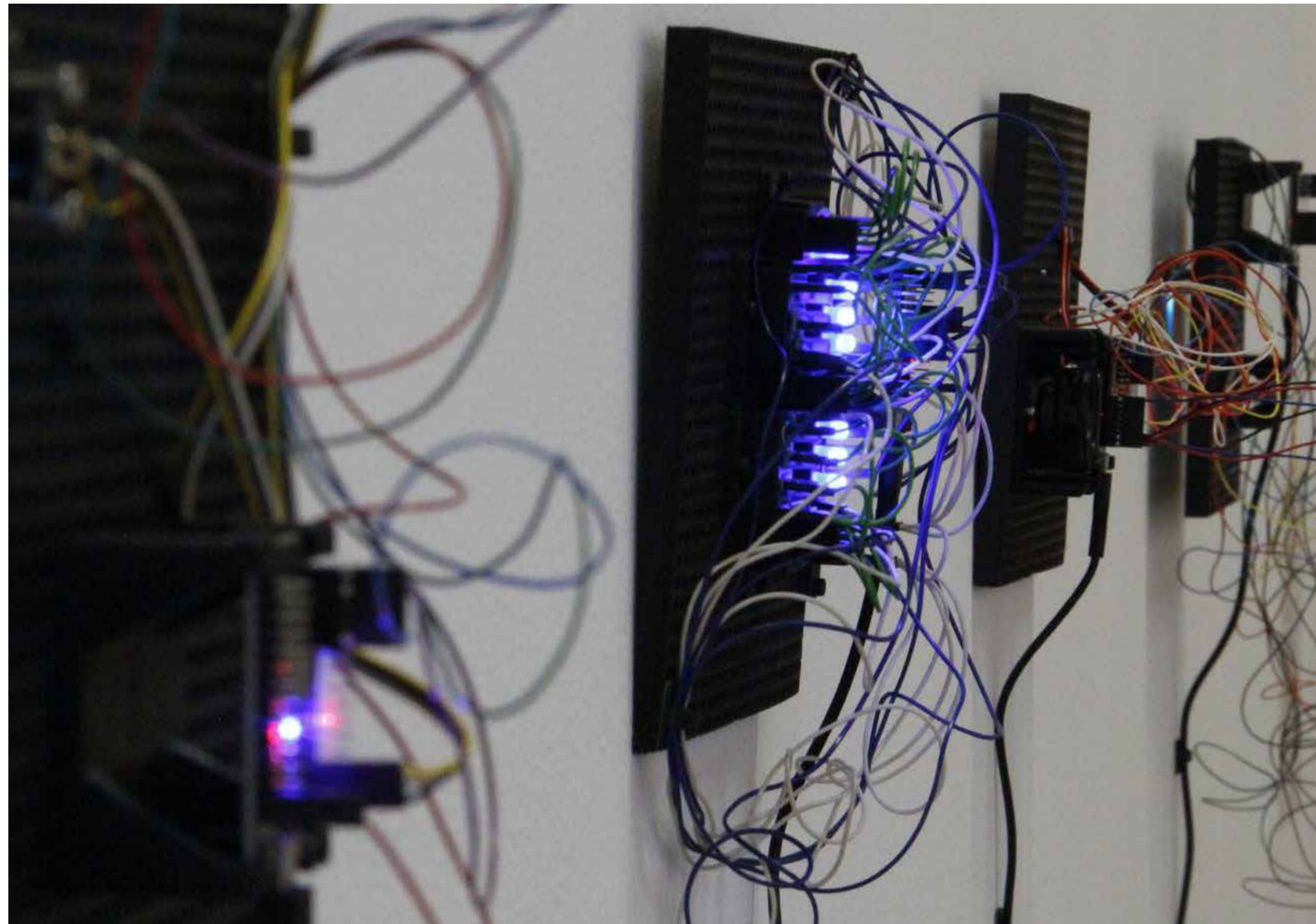


Binary Landscapes

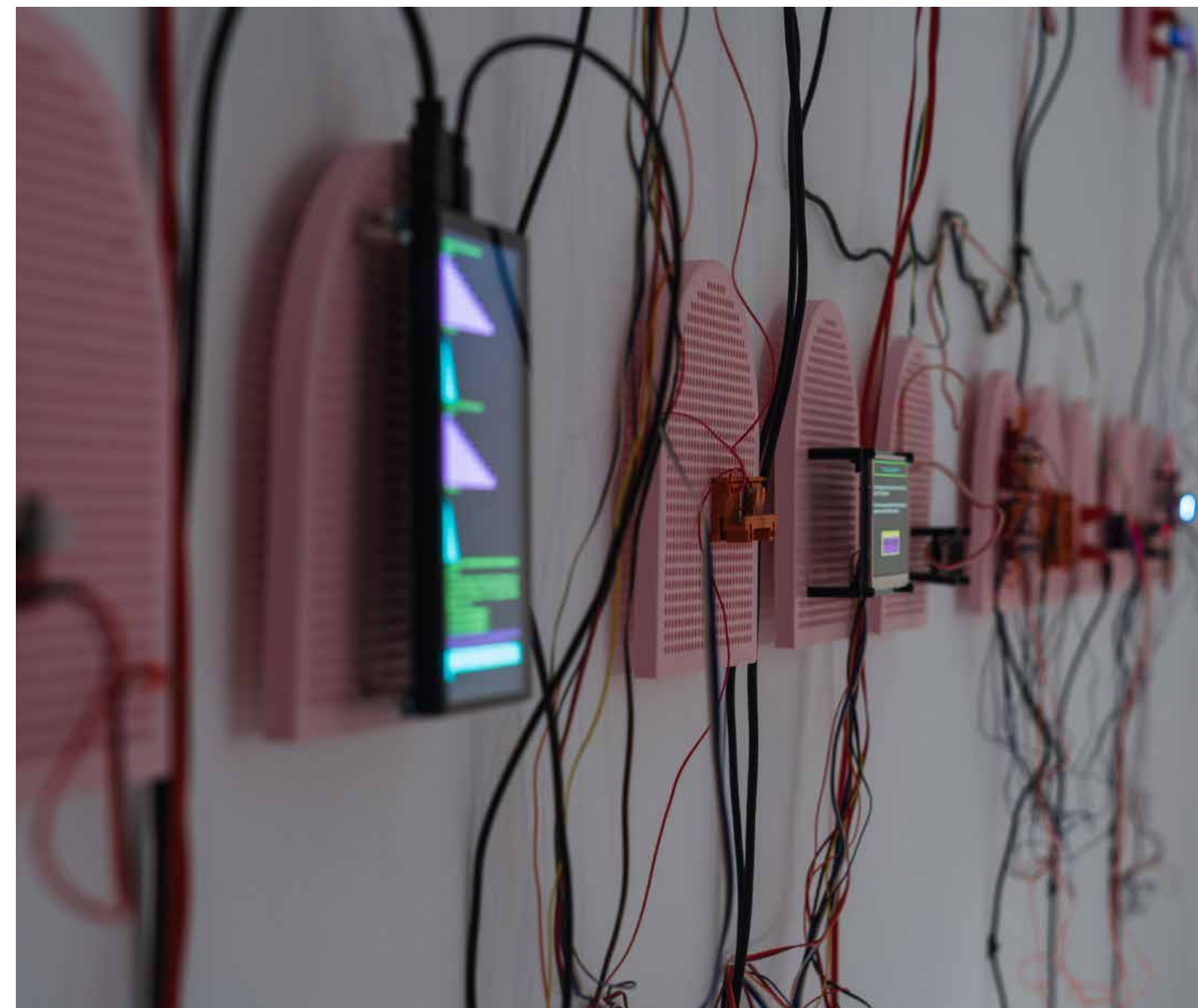
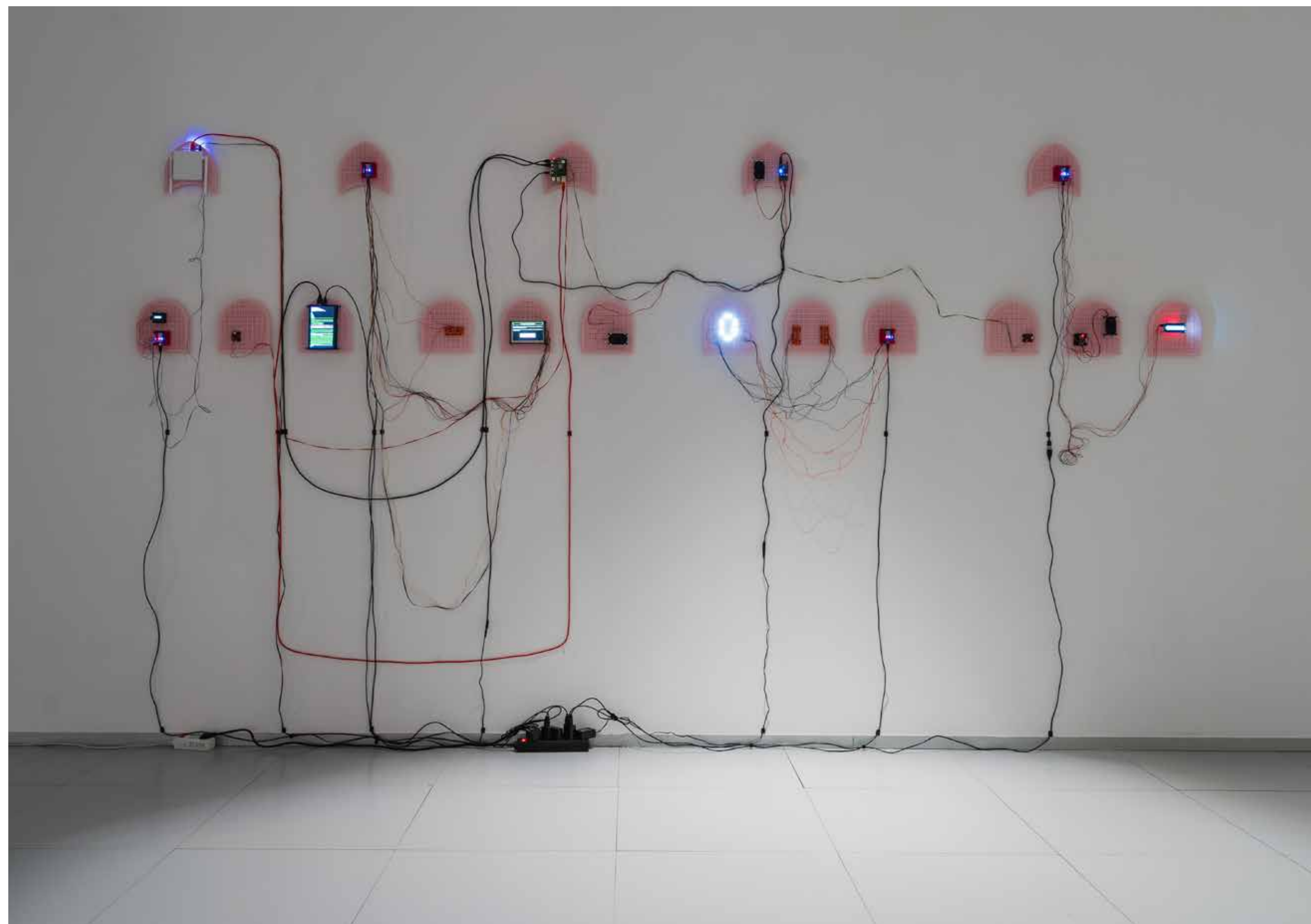
#Woman Critical Interface, 2021-2023



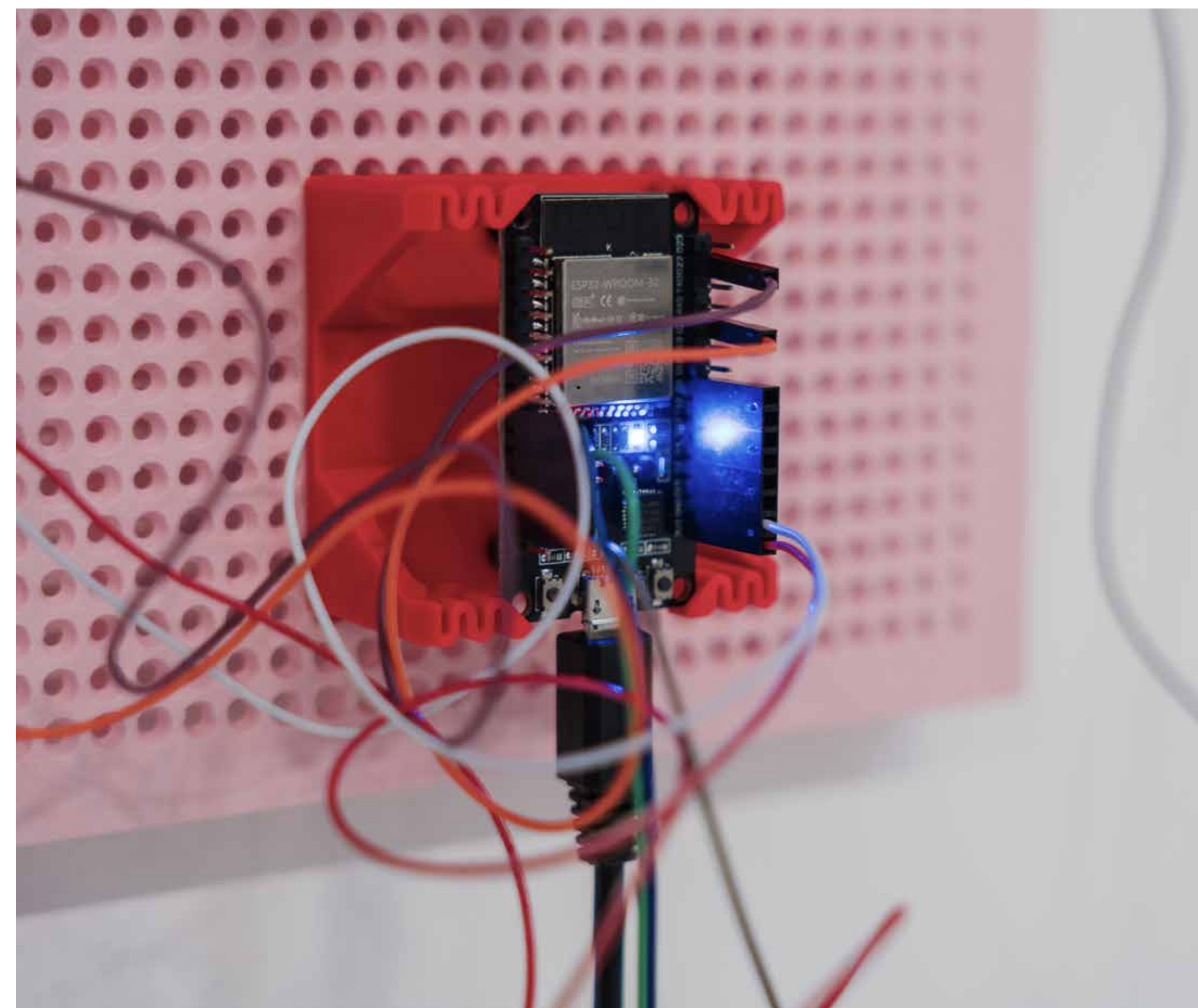
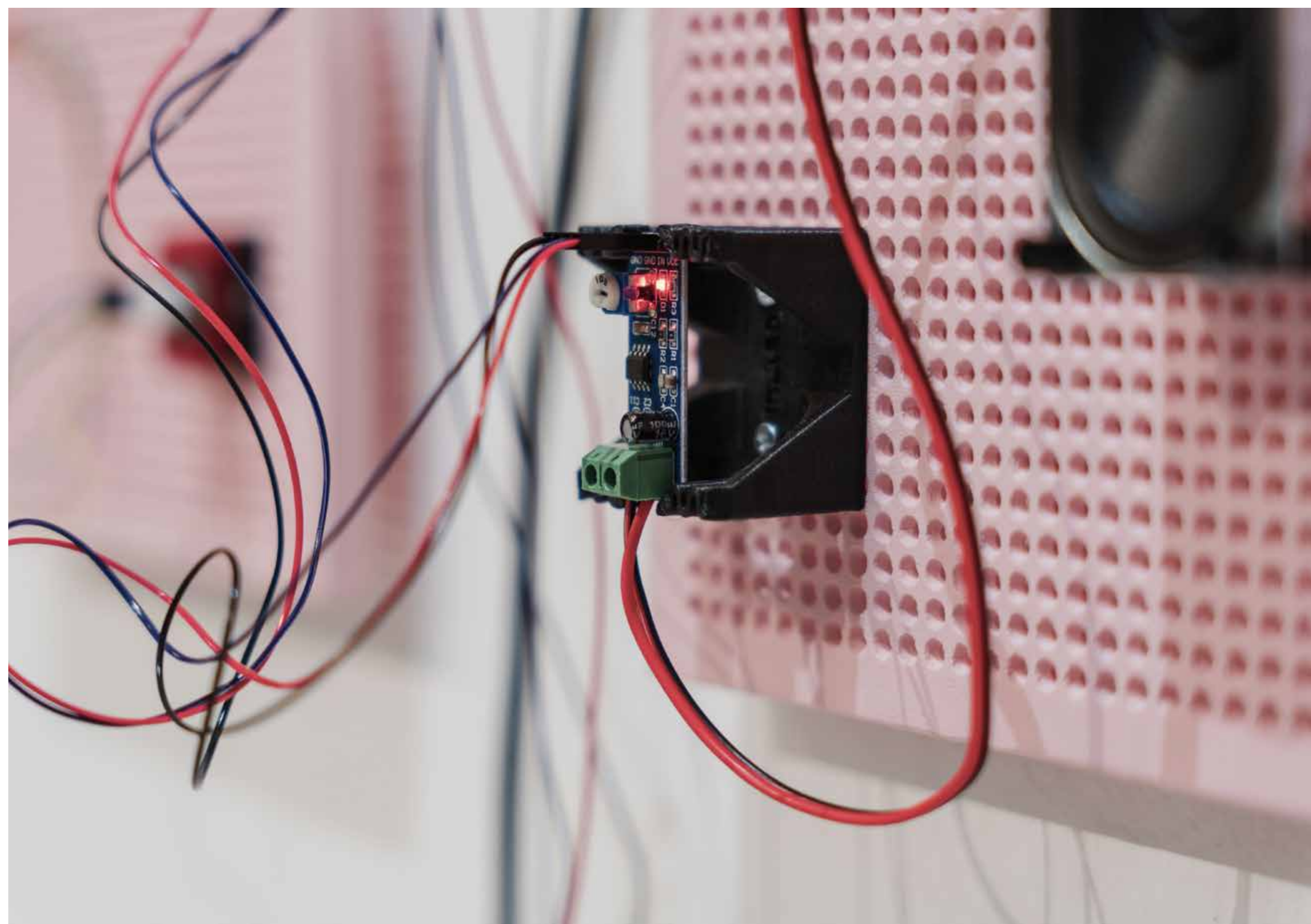
Conversatorio, 2023



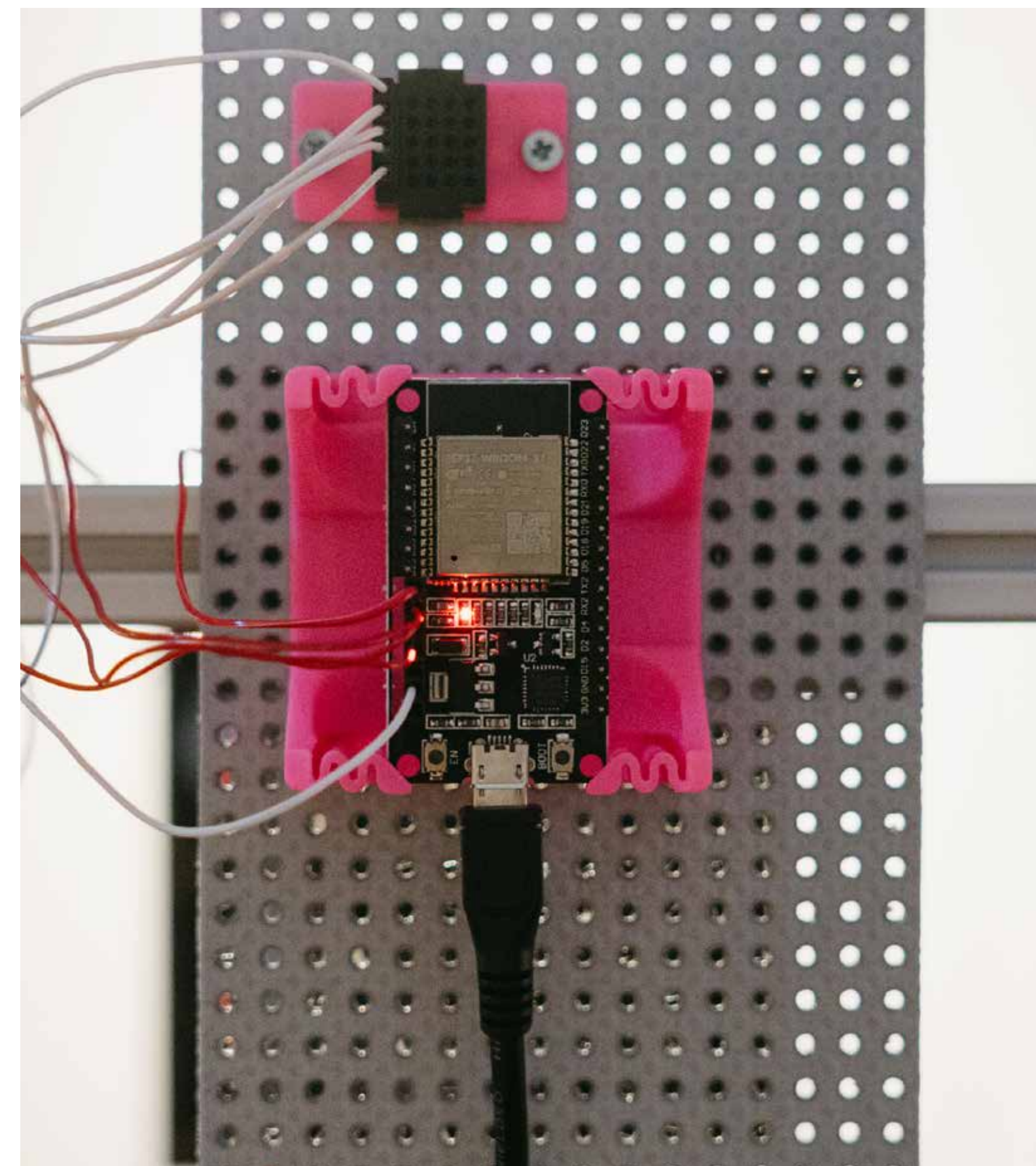
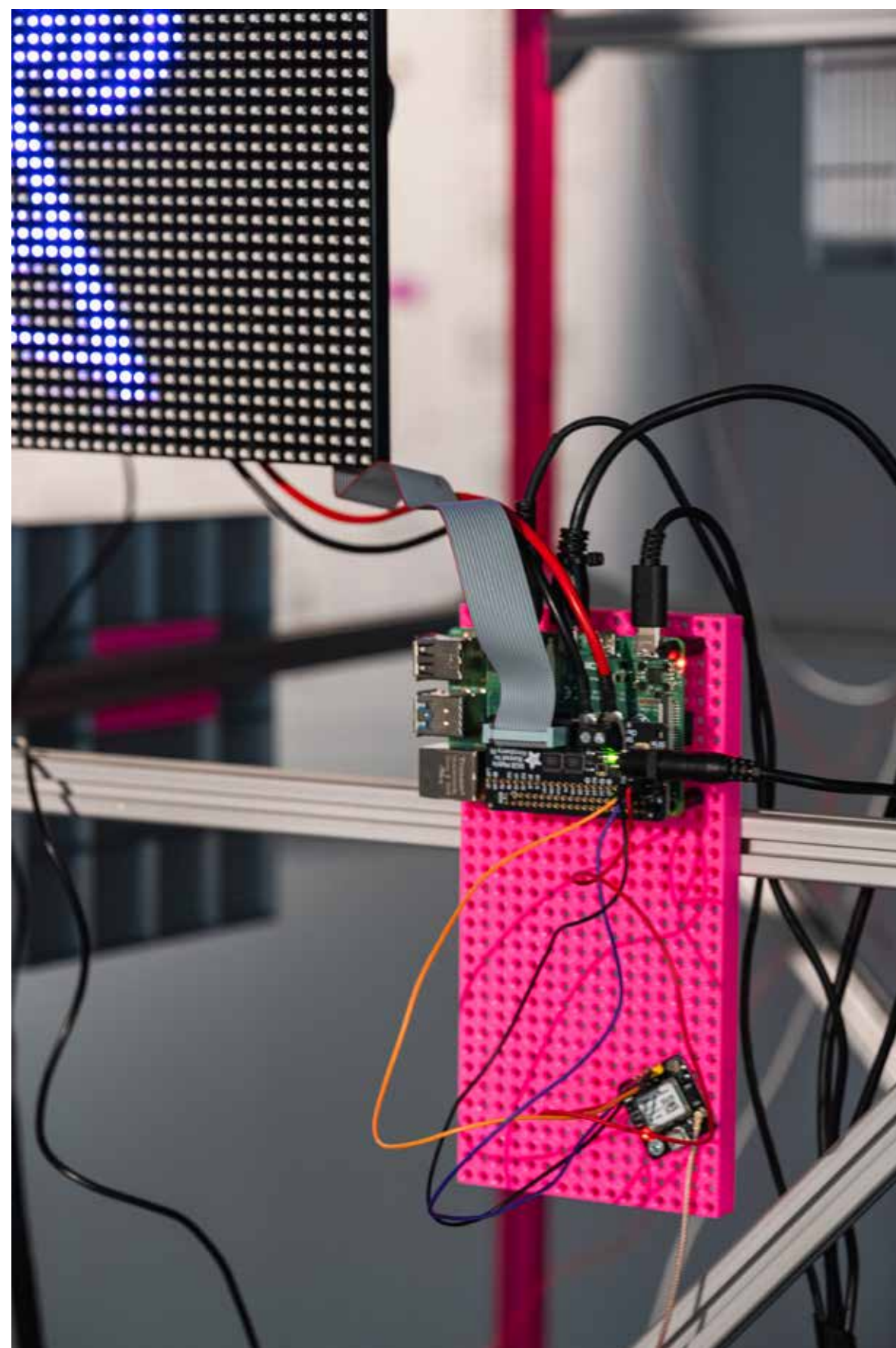
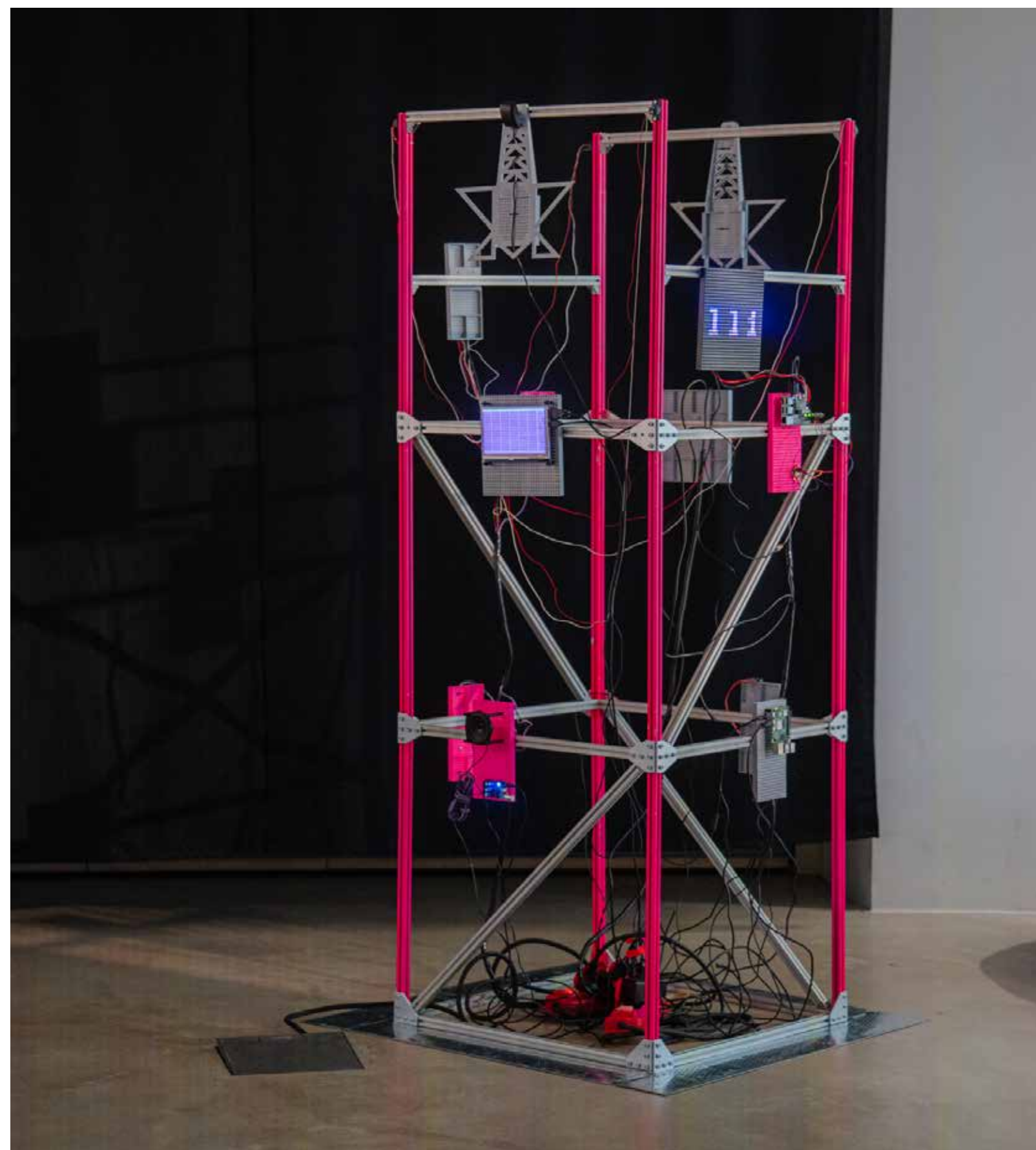
Hash Breakdown, 2024



Hash Breakdown, 2024



Stratum, 2025



offDAC community (since 2023)

offdac.com

15/11/23 16:00
Etopia (Aula 3) Zaragoza
offDAC event 001
FREE ENTRY

001. GENDER, TECHNOLOGY and POWER NETWORKS

Azahara Cerezo
Paula Nishijima
Silvia Bindas
in conversation

ARTeCHÓ
ETOPIA
Co-funded by the European Union
offDAC



13/01/24 11:00 - 17:00
offDAC LAB Košická 52/A, Bratislava
offDAC event 002
Live Coding Workshop

Live Coding

Tidal Cycles workshop



Live Coding workshop organized by offDAC with kind support from Tidal Cycles. Registration is necessary, see link in the QR code below. No previous knowledge is needed. Participation is free of charge. Snacks and beverages will be provided by offDAC. For further information see offdac.com or write us at info@offdac.com

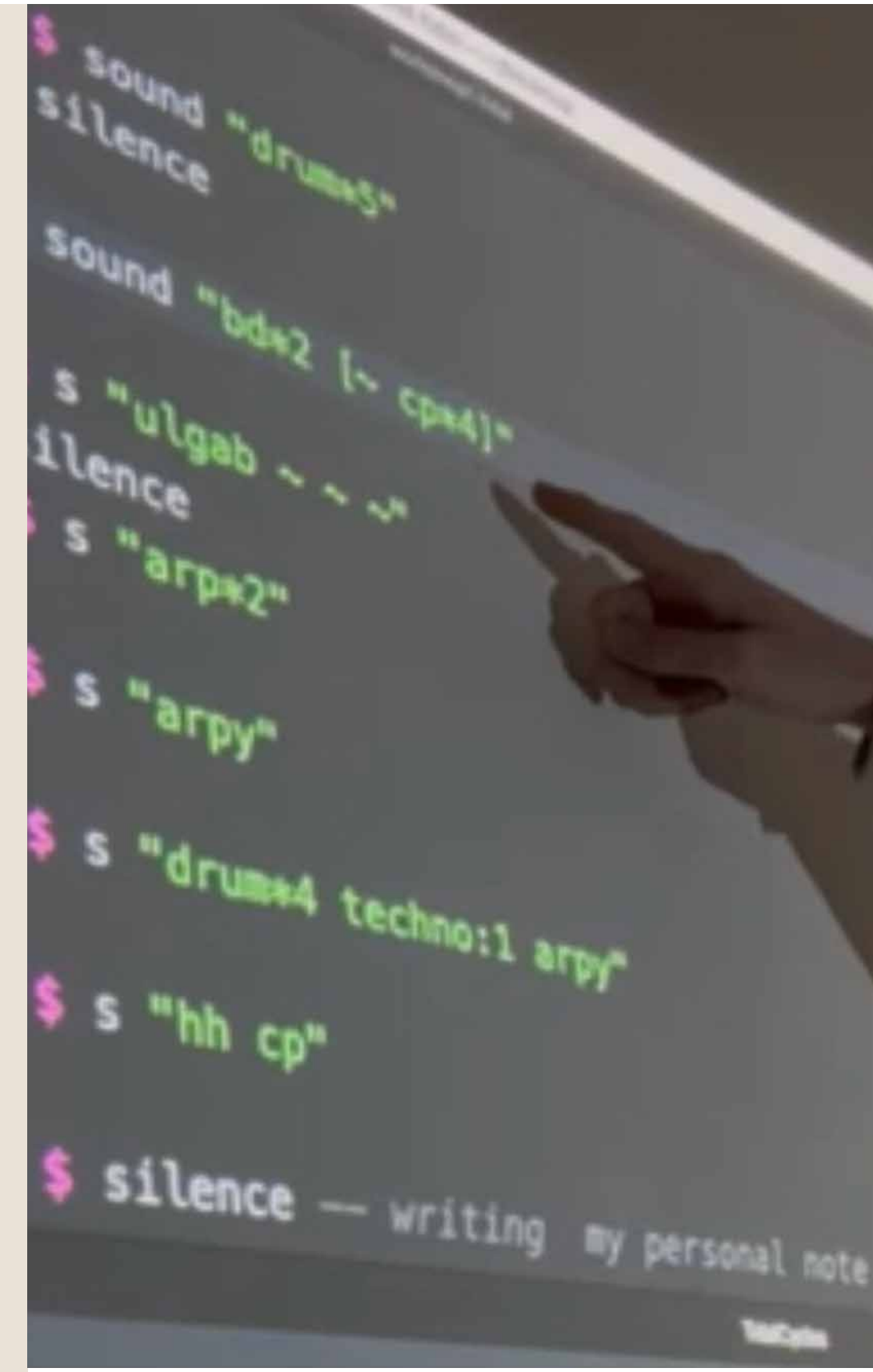
Organized by:



With kind support by:



Register:



OPEN HARDWARE PRINCIPLES

<https://opencourse.com>

”Open hardware,” or “open source hardware,” refers to the design specifications of a physical object which are licensed in such a way that said object can be studied, modified, created, and distributed by anyone.

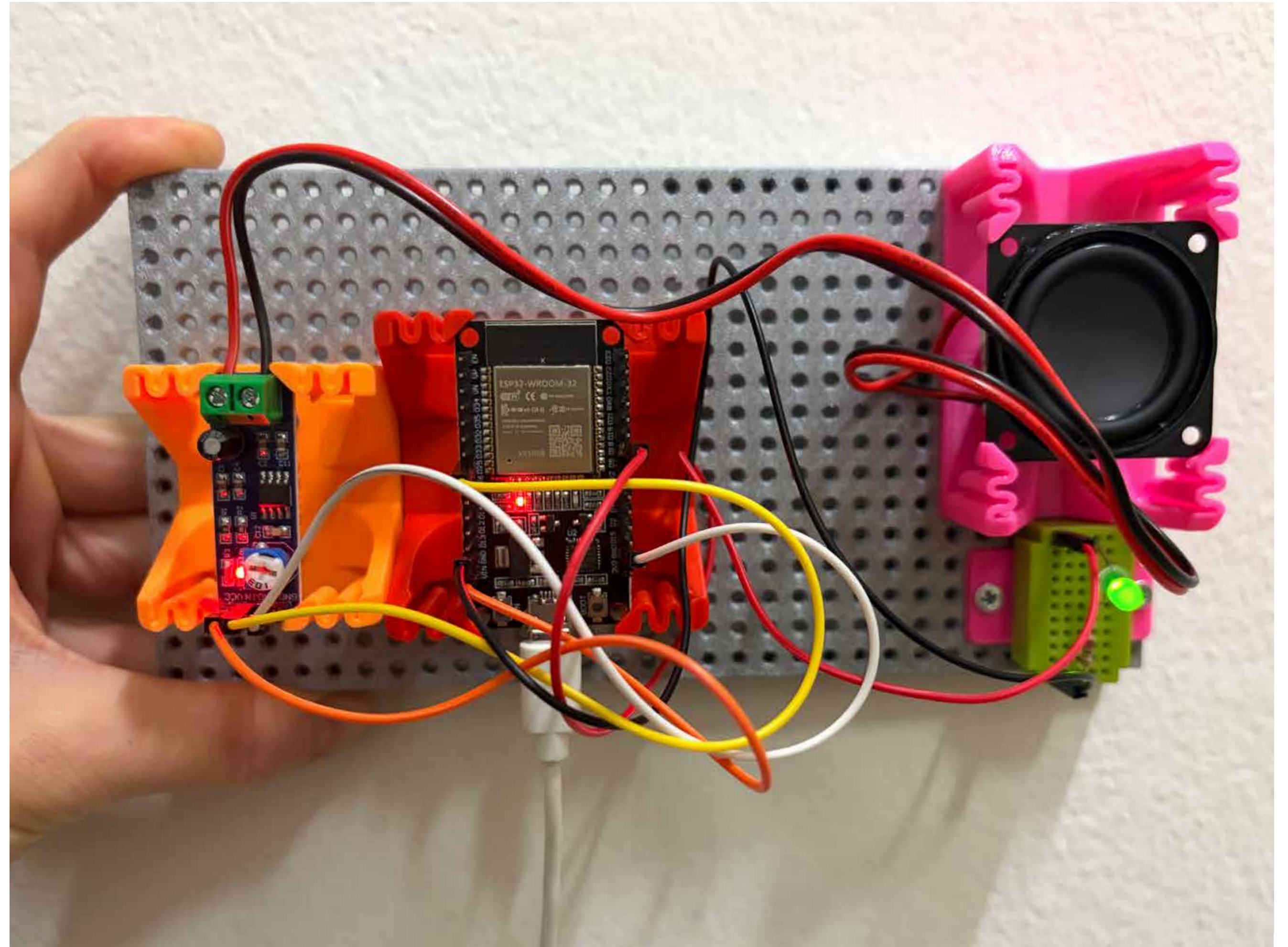
OPEN HARDWARE PRINCIPLES

<https://opencourse.com>

- the “source code” for open hardware—schematics, blueprints, logic designs, code, etc.—is available for modification by anyone (under permissive licenses, such as MIT license)**
- to provide as many people as possible the ability to construct, remix, and share their knowledge of hardware design and function**

PLAN FOR TODAY IS:

**MAKE OUR
OWN
CREATIVE
HARDWARE
ARTEFACT**



PLEASE RAISE YOUR HAND WHO:

- EVER SOLDERED SOMETHING

PLEASE RAISE YOUR HAND WHO:

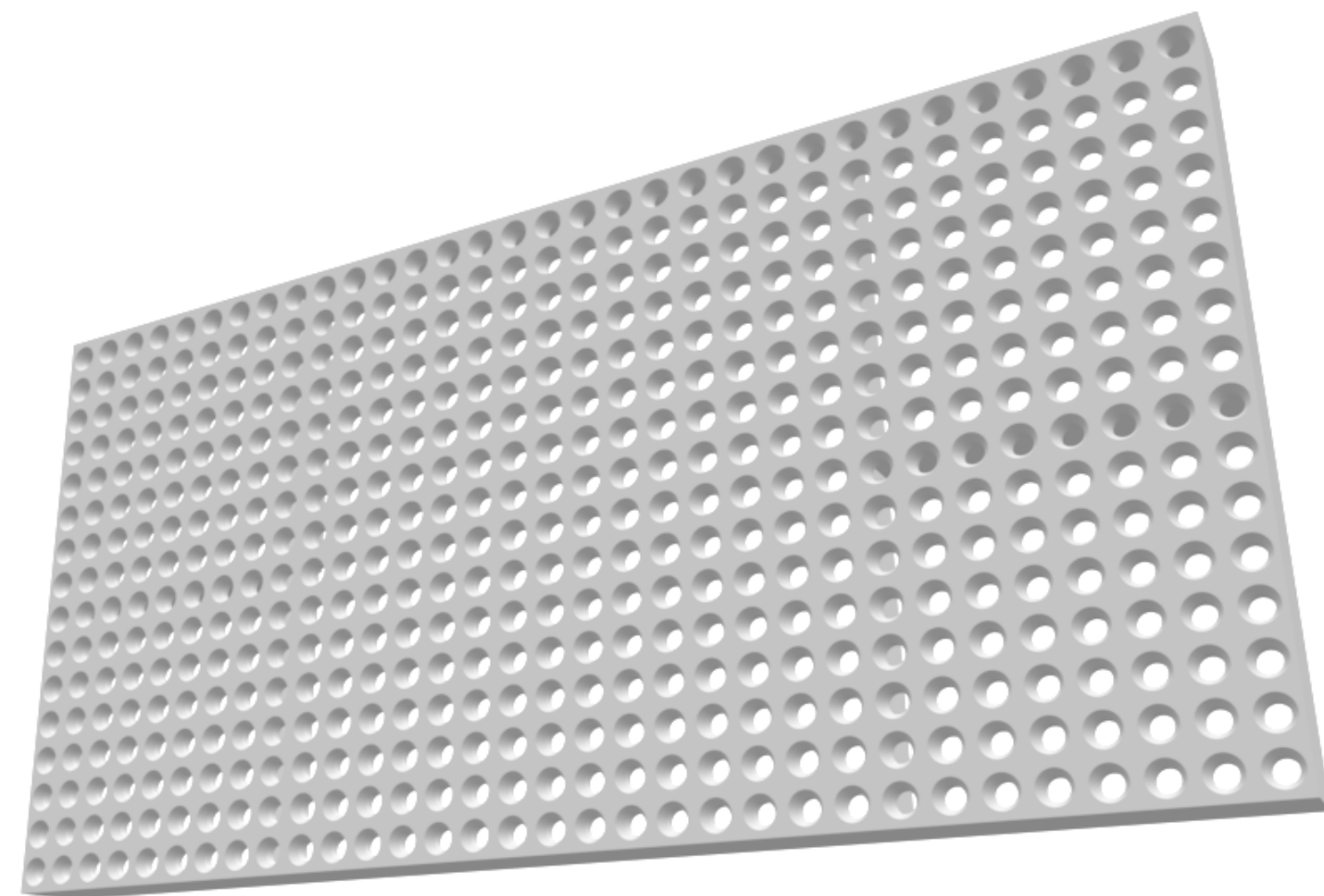
- EVER SOLDERED SOMETHING**
- KNOWS WHAT IS AN “ESP32”**

PLEASE RAISE YOUR HAND WHO:

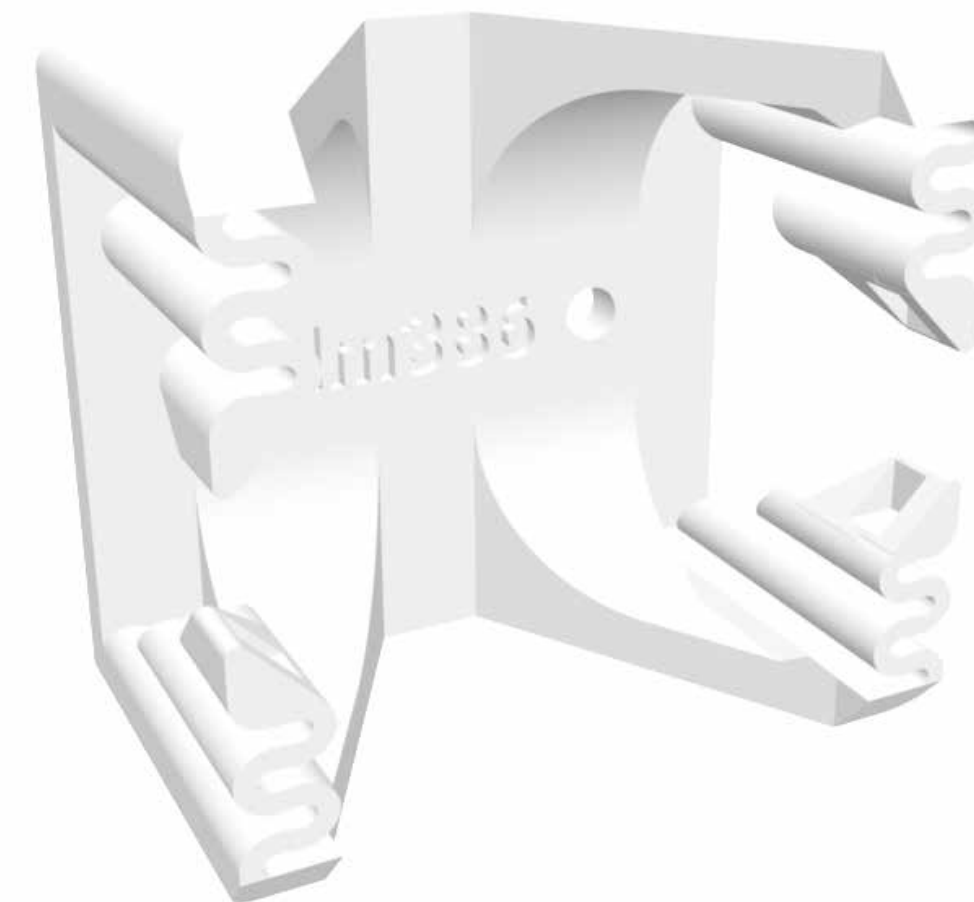
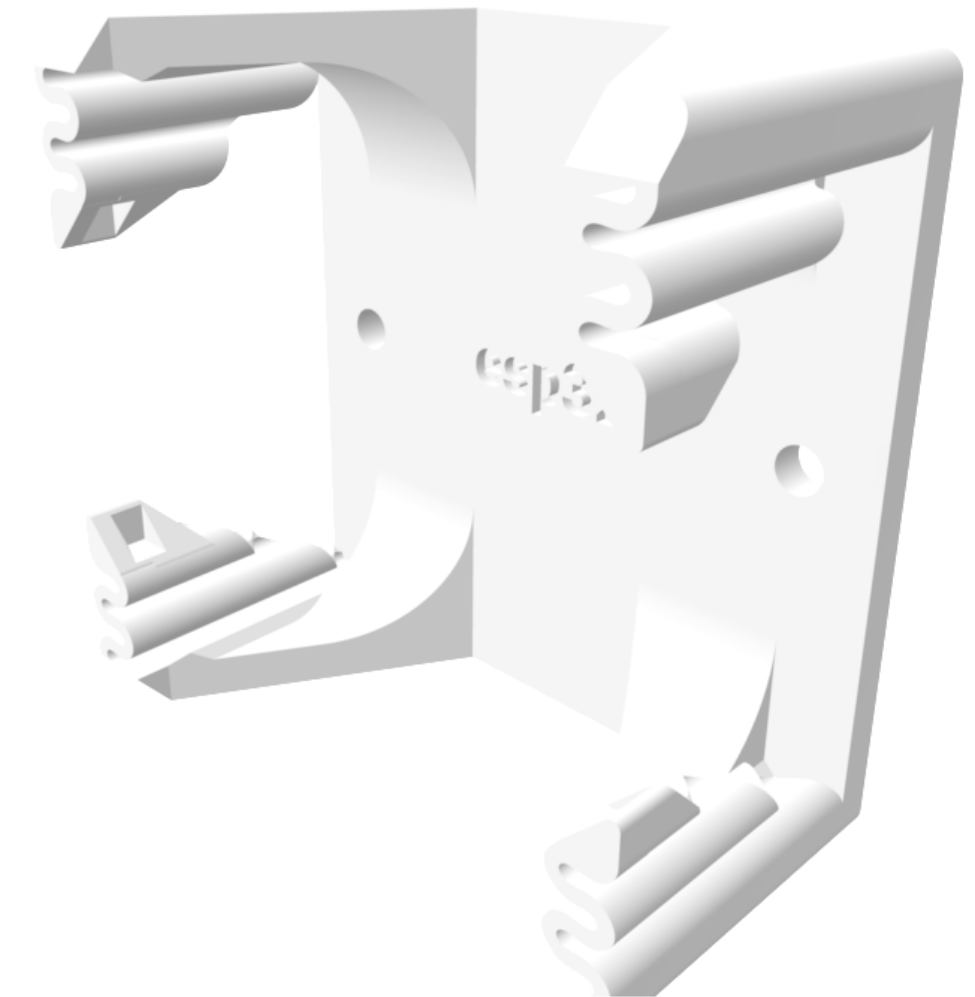
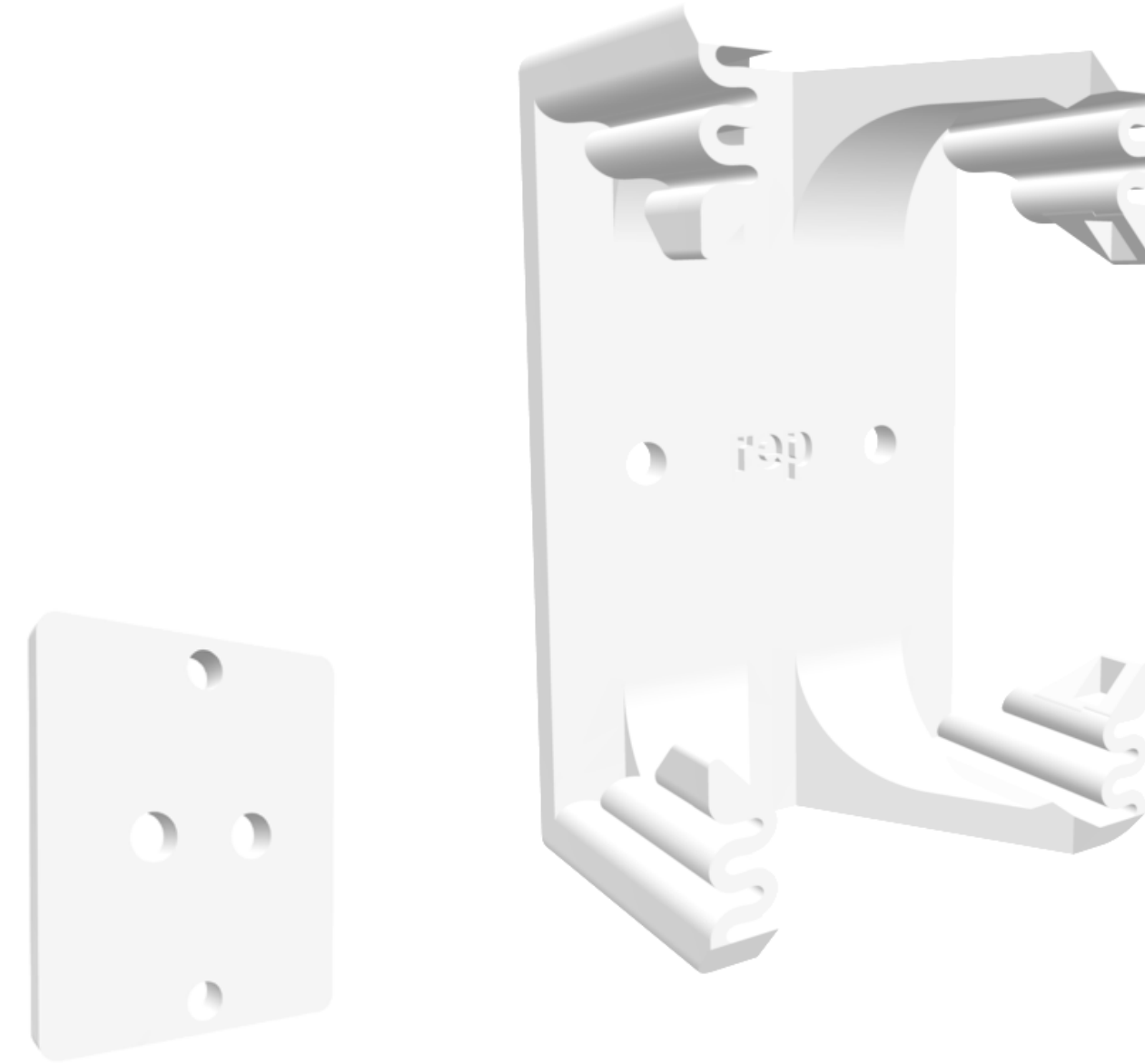
- EVER SOLDERED SOMETHING**
- KNOWS WHAT IS AN “ESP32”**
- EVER CODED IN PYTHON**

Items we will use:

1) 3D prints



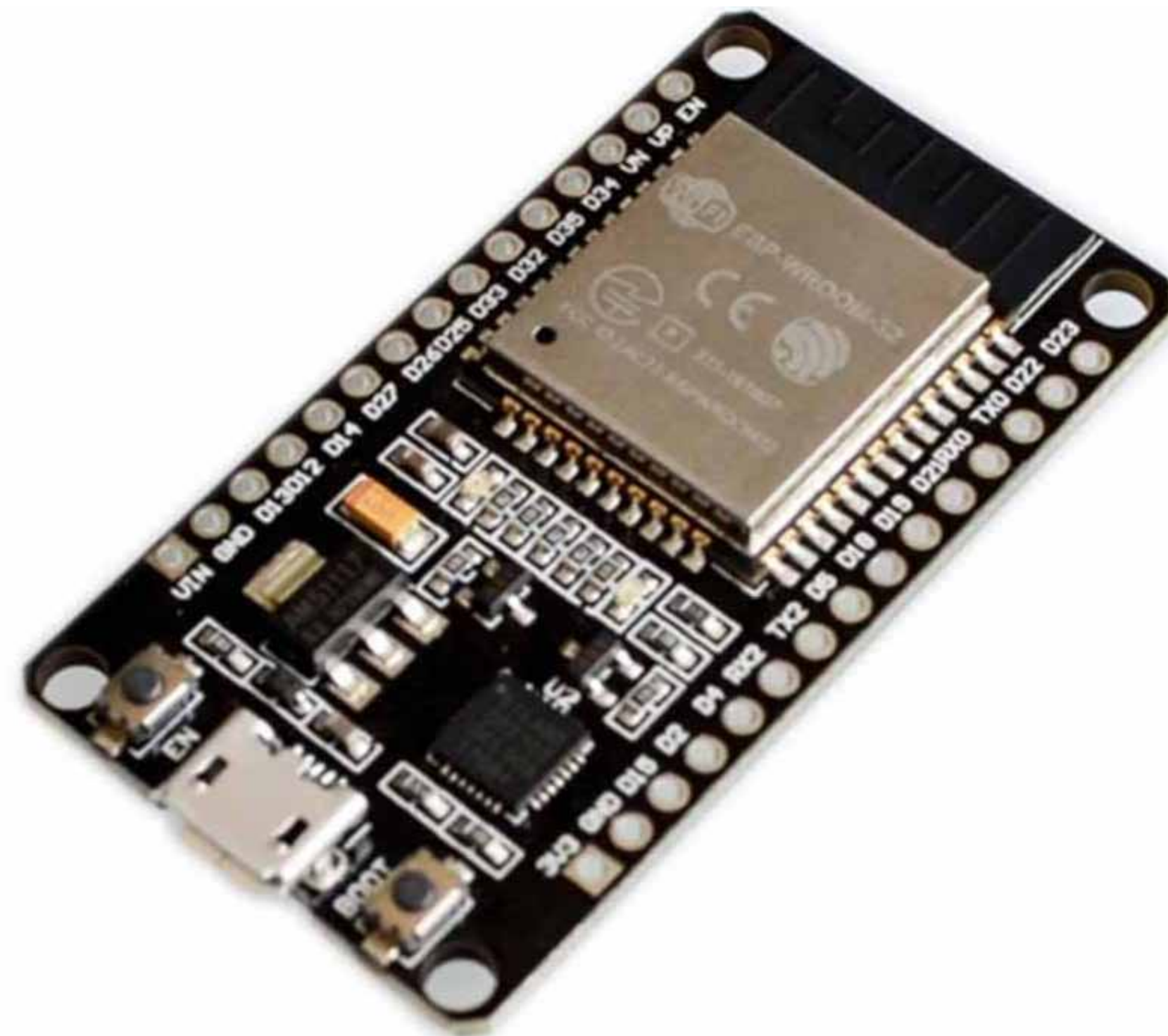
BASE



HOLDERS

Items we will use:

2) ESP32 microcontroller



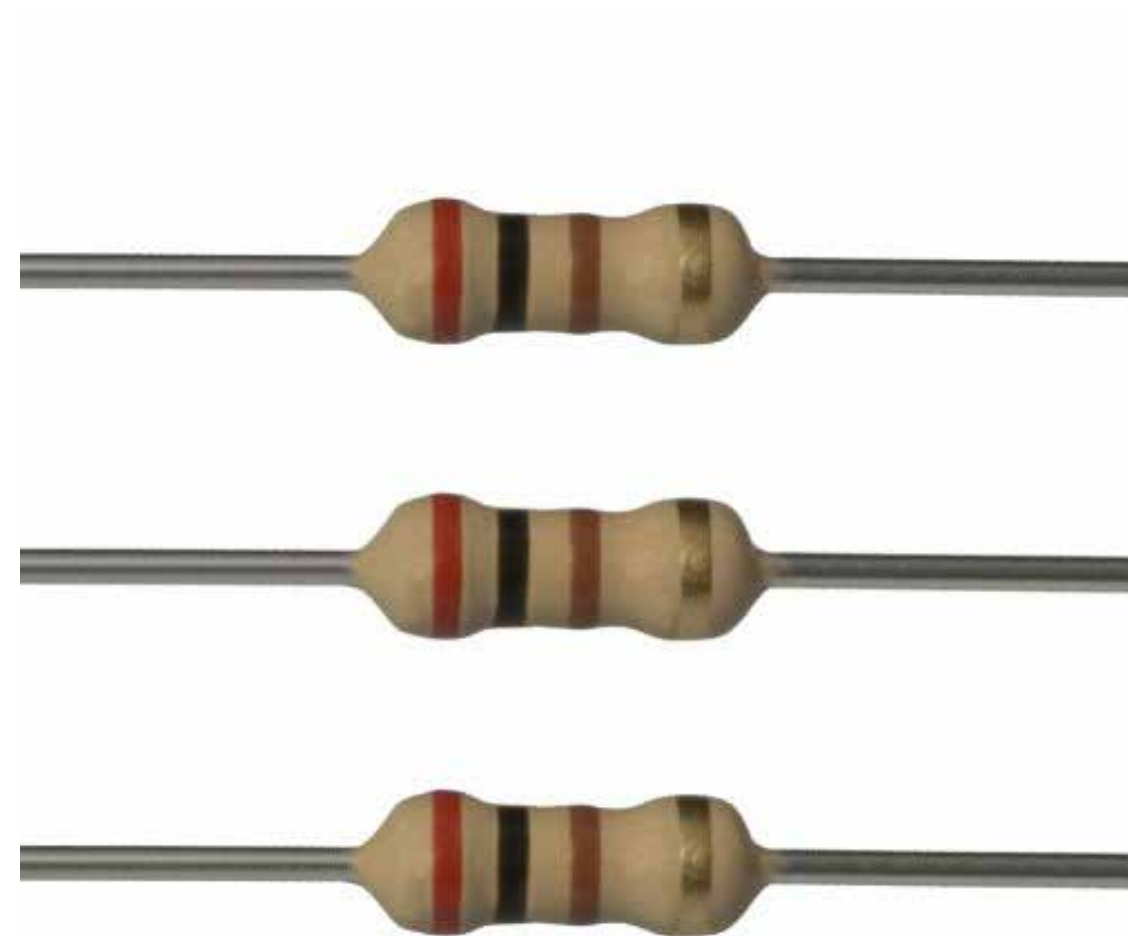
3) LED (Light Emitting Diode)



**green
red
yellow
white**

Items we will use:

4) resistor (200Ω)



**to regulate
the flow of
electrical
current in
a circuit**

5) mini breadboard



Items we will use:

6) LM386 amplifier



7) mini speaker (4Ω 3W)



Items we will use:

8) wires, M3 screws, tools



Design your board

1) Choose 1 base, 1 holder from each type, 1 wall mount

2) Choose one breadboard and connect it to the holder (it has to make “click”, adjust the holder holes if needed)

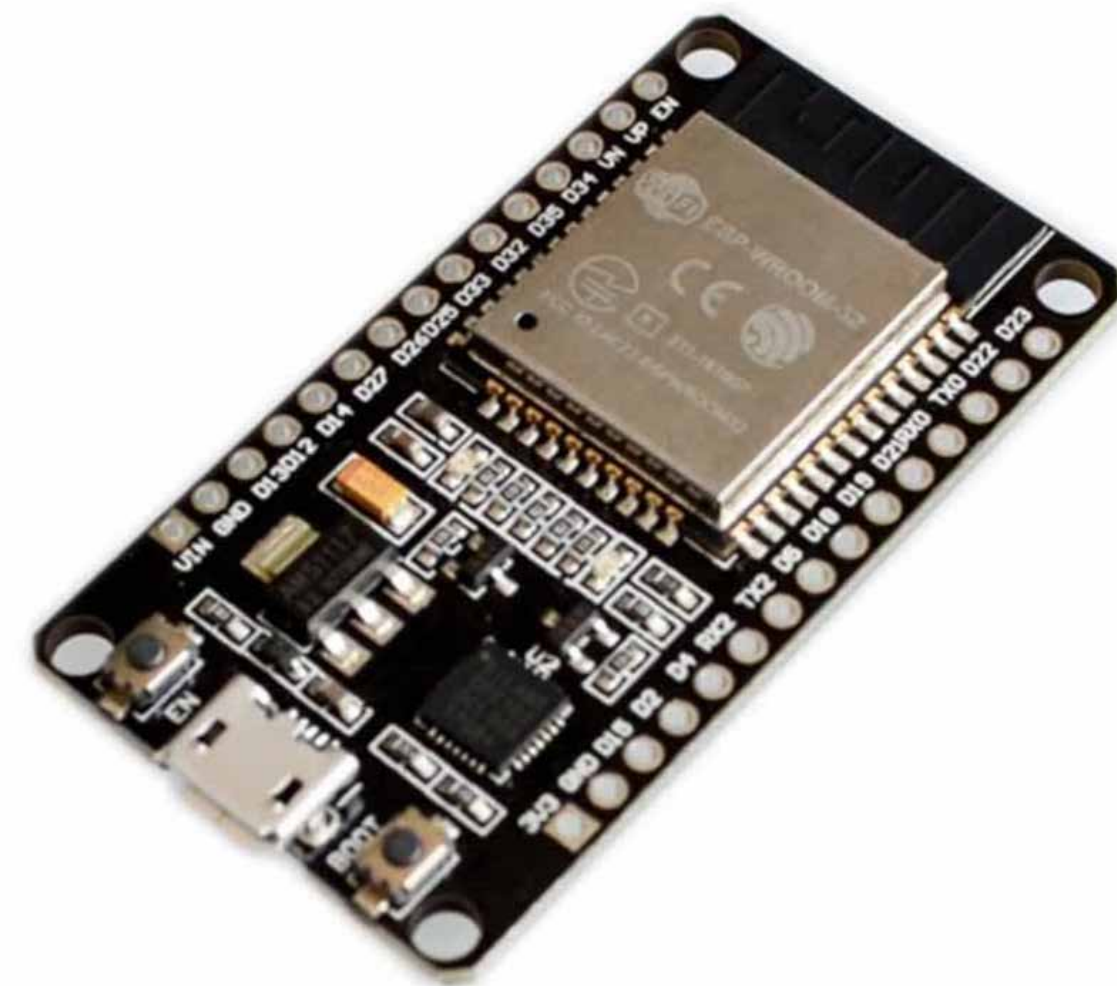
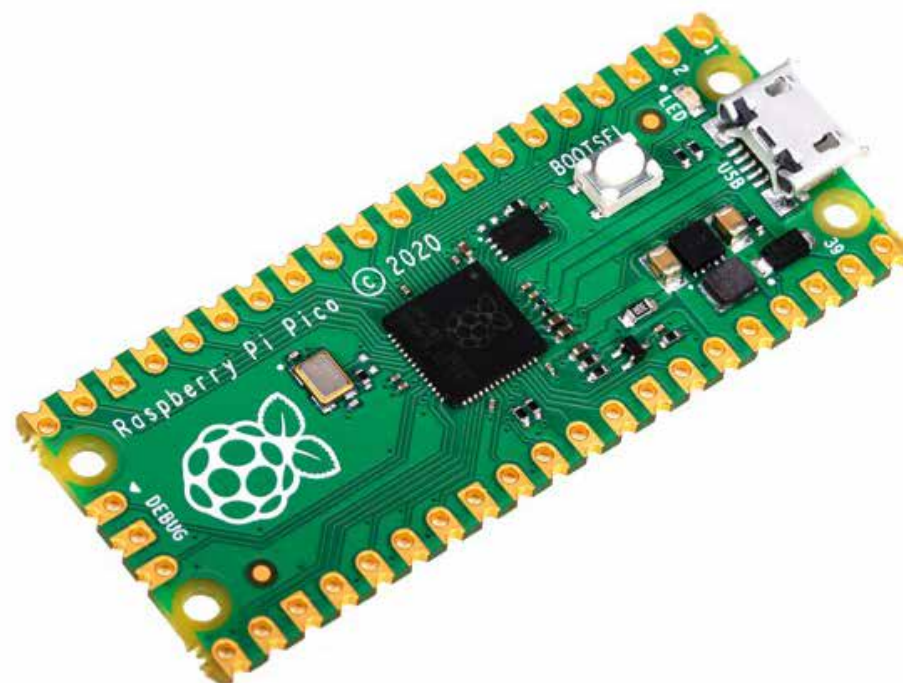
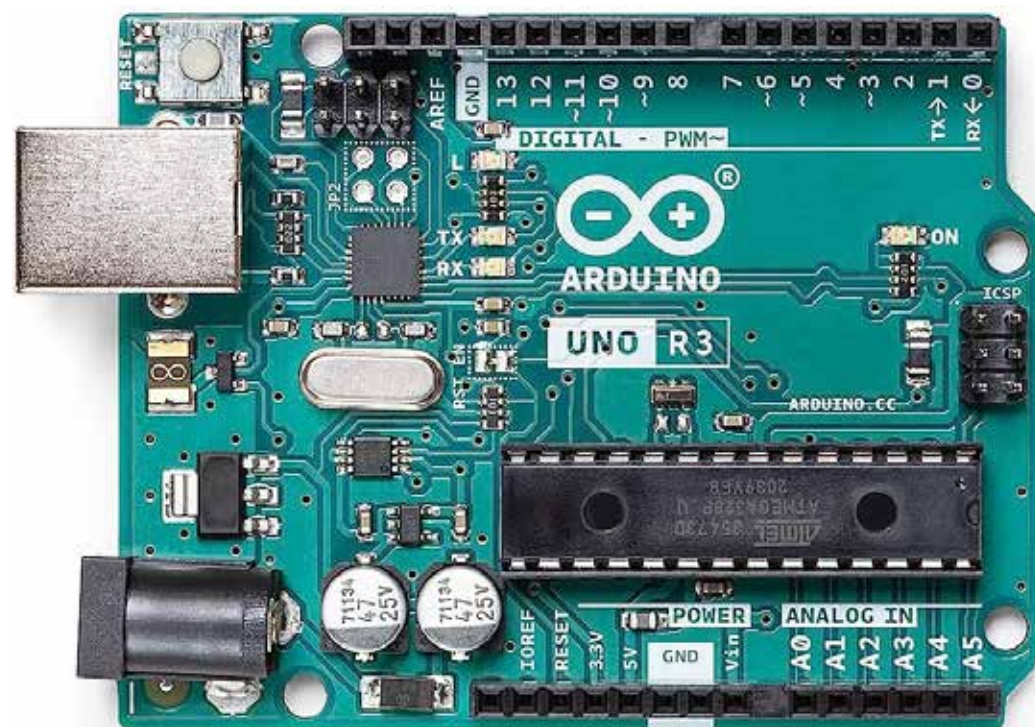
3) Locate the items on the board and screw them in the holes



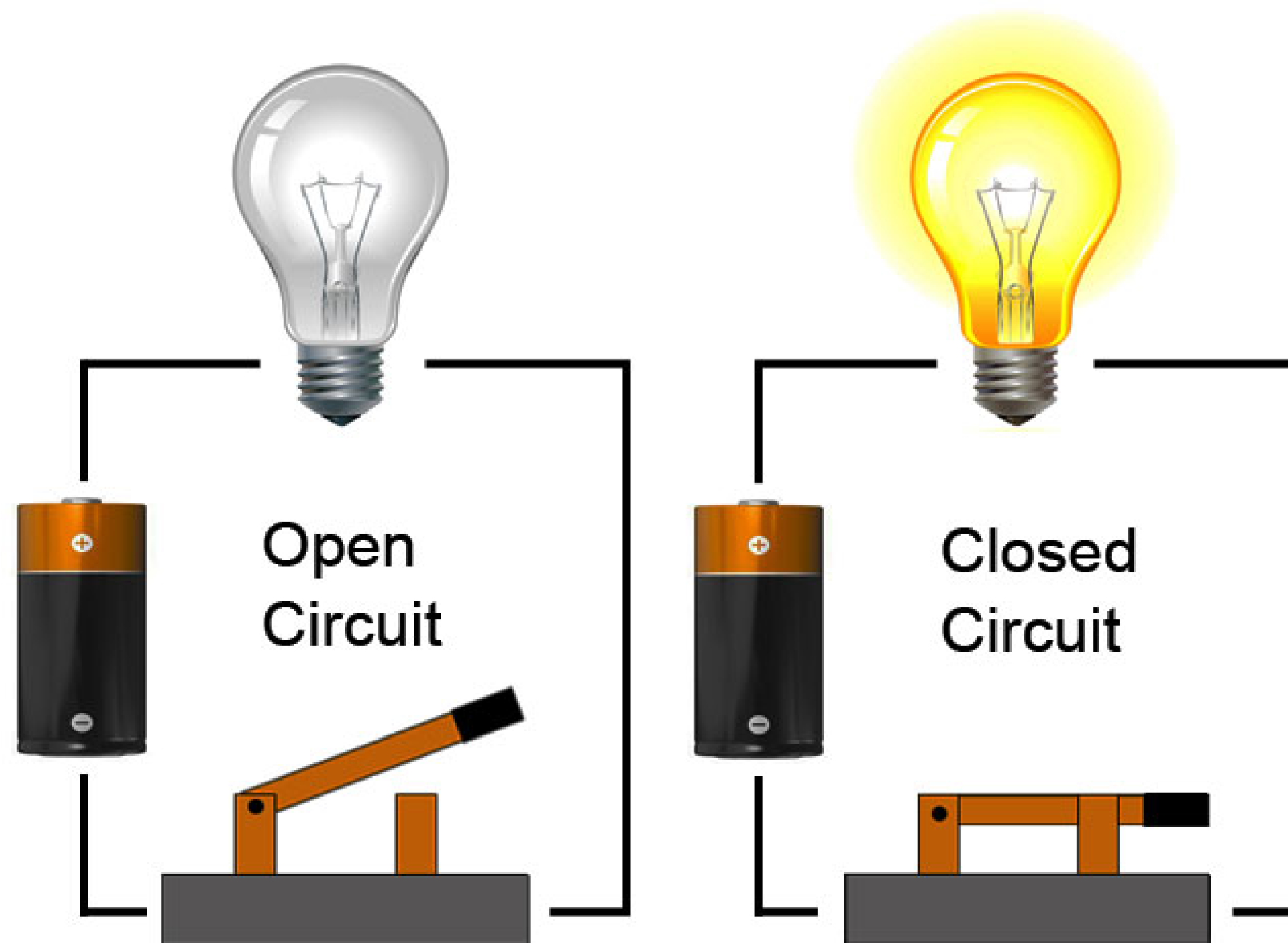
brief introduction to electronics:

MICROCONTROLLERS

- = a small, low-cost computer-on-a-chip
- designed to perform a specific set of tasks
- applications: controlling machines, sensing and monitoring devices, art, IoT
- typically consists of a central processing unit (CPU), memory, input/output (I/O) pin
- common examples = Arduino, Raspberry Pi Pico, ESP 32



OPEN AND CLOSED CIRCUITS



circuit is a complete and closed path through which electric current can flow

closed circuit would allow the flow of electricity between **power (+)** and ground (-)

open circuit would break the flow of electricity between **power (+)** and ground (-)

POSITIVE VOLTAGE AND GROUND

Positive Voltage:

“VCC”

“3V3”

“+”

Ground (carries a voltage of 0V):

“GND”

“-”

!!!! It is very important to prevent short circuits by making sure that the **positive voltage (+)** is never wired directly to ground (-) !!!!

INSPECT THE COMPONENTS AND FIND POSITIVE VOLTAGE AND GROUND ON EACH

DIGITAL MULTIMETER

= a device used to measure:

- electric current (amps)
- voltage (volts)
- resistance (ohms)

= great for troubleshooting

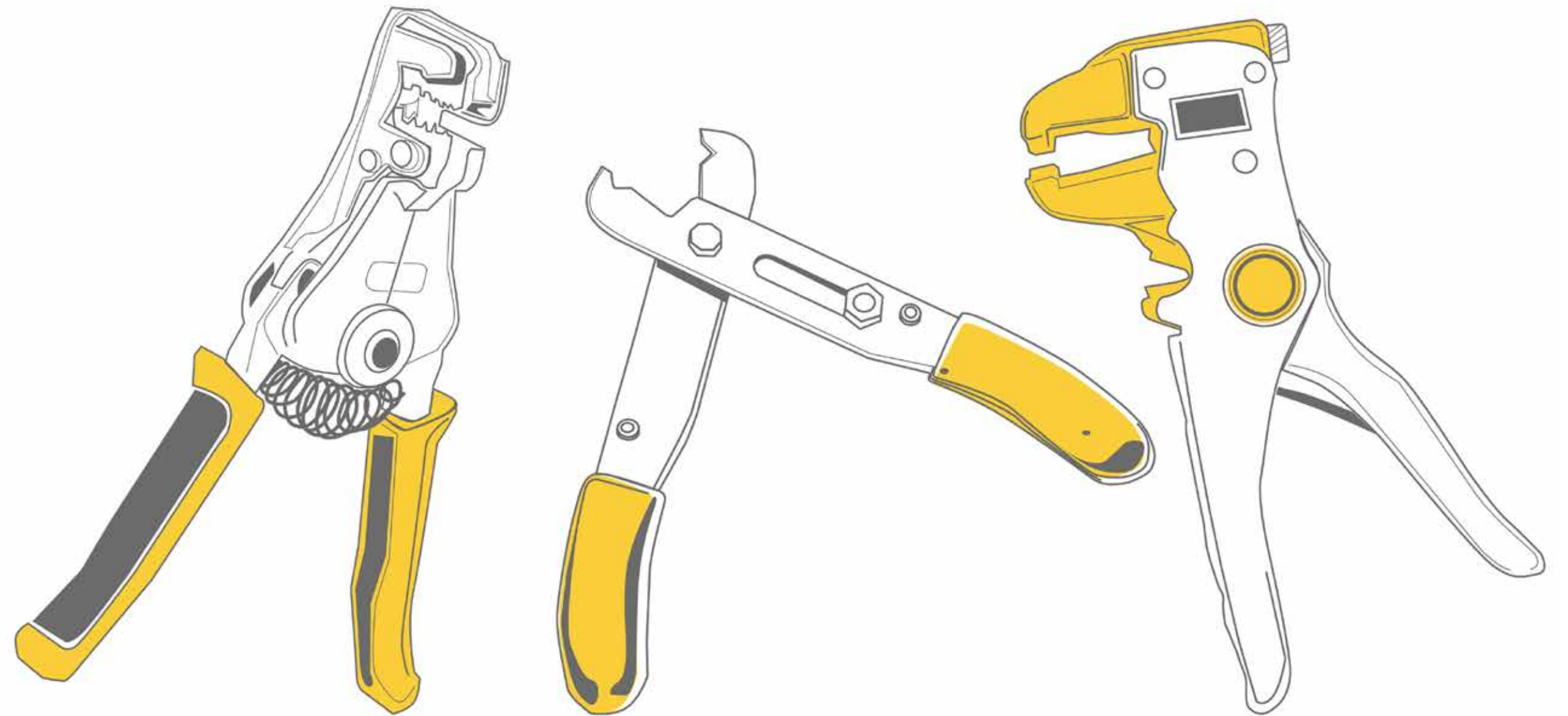
= capable of measuring both
AC and DC voltage



WIRE CUTTERS & STRIPPERS

= designing your preferred cable length

= essential for stripping off the insulation layer of the wire



HELPING 3rd HAND

= for holding circuit boards or wire when soldering or tinning



Soldering the PINs on ESP32

Explanatory video: <https://www.youtube.com/shorts/kY99mjw638A>

solder joint check:

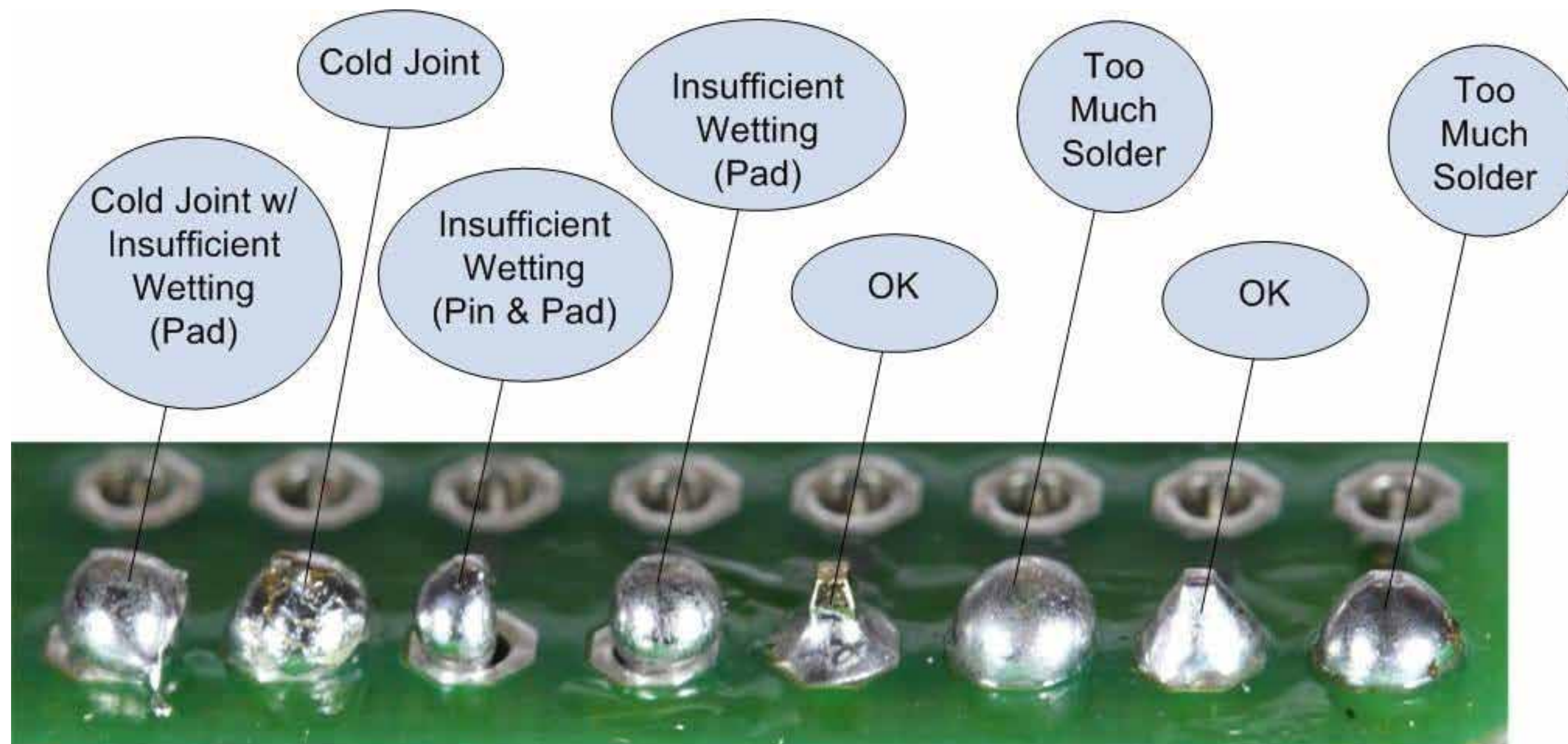
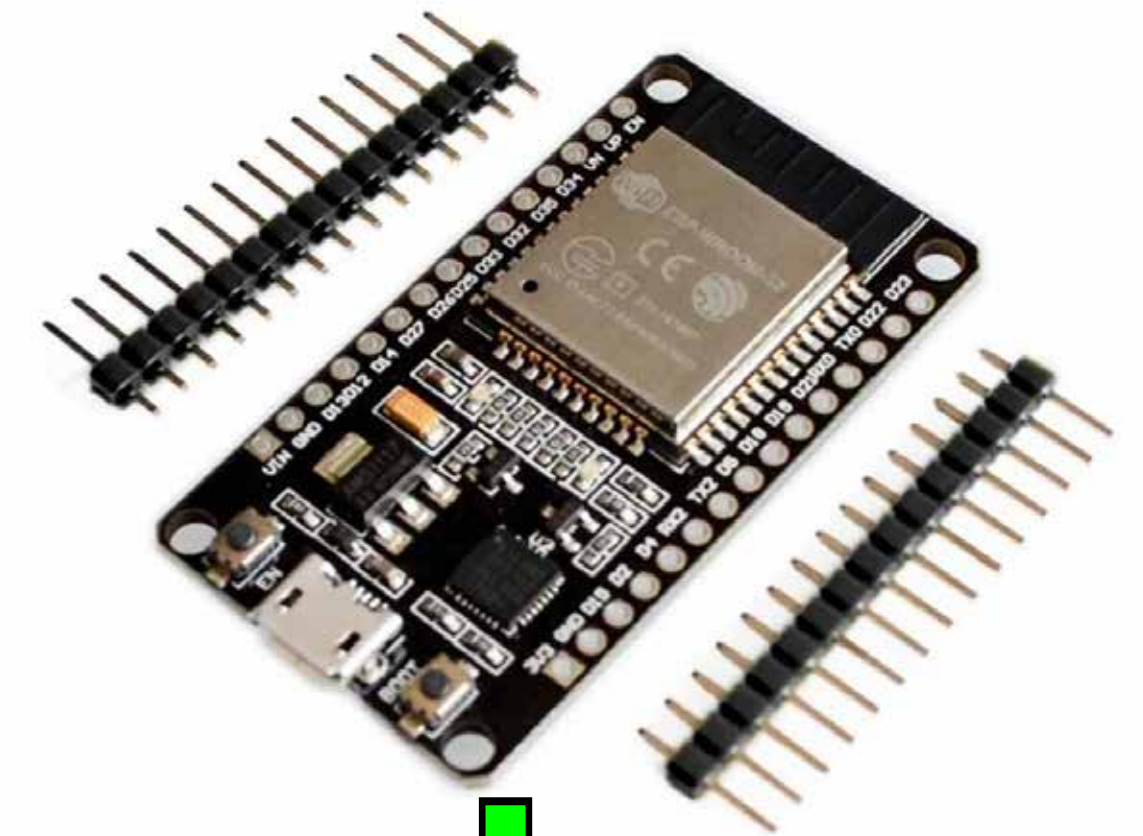
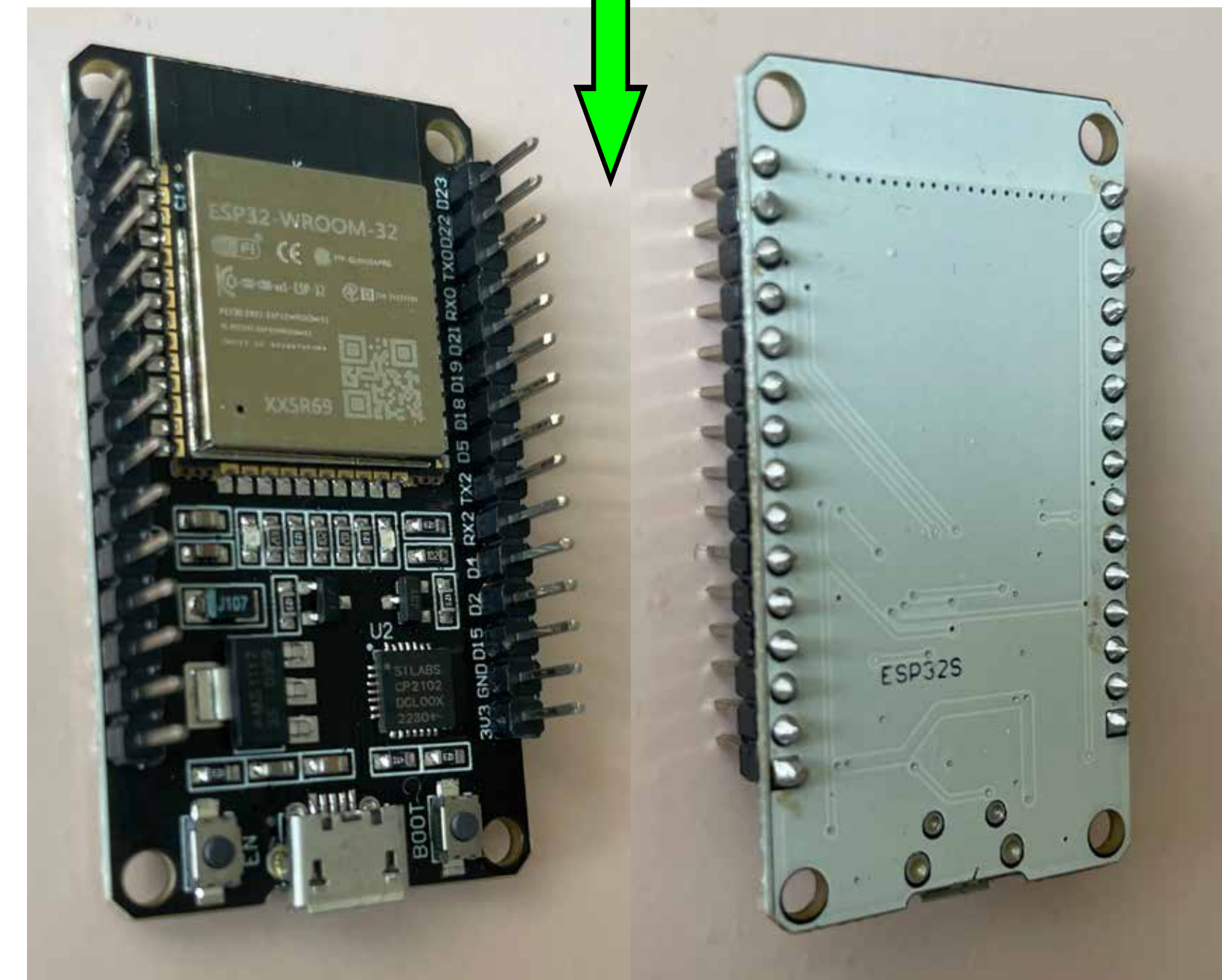
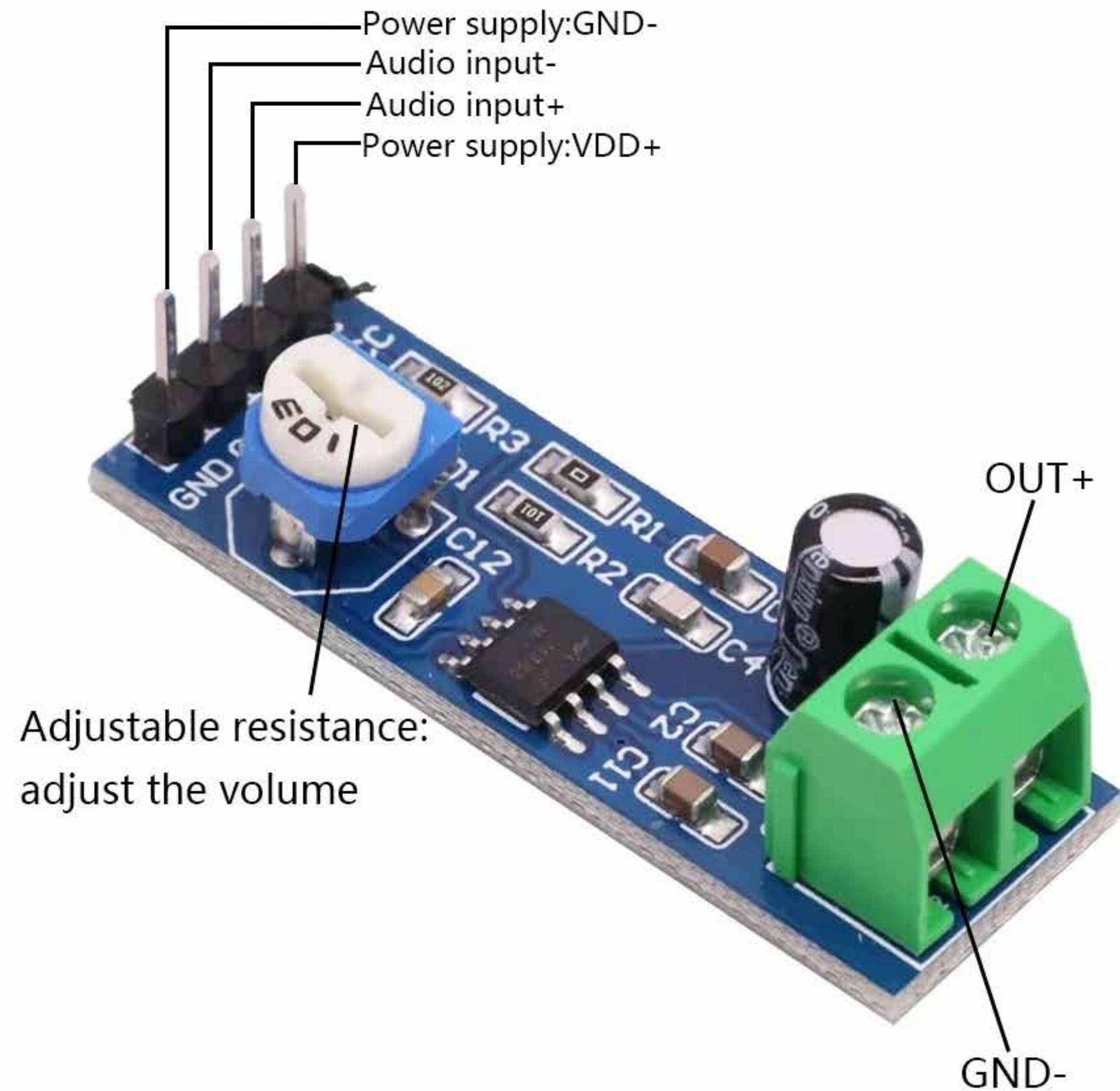


Image source: <https://learn.adafruit.com/adafruit-guide-excellent-soldering/common-problems>

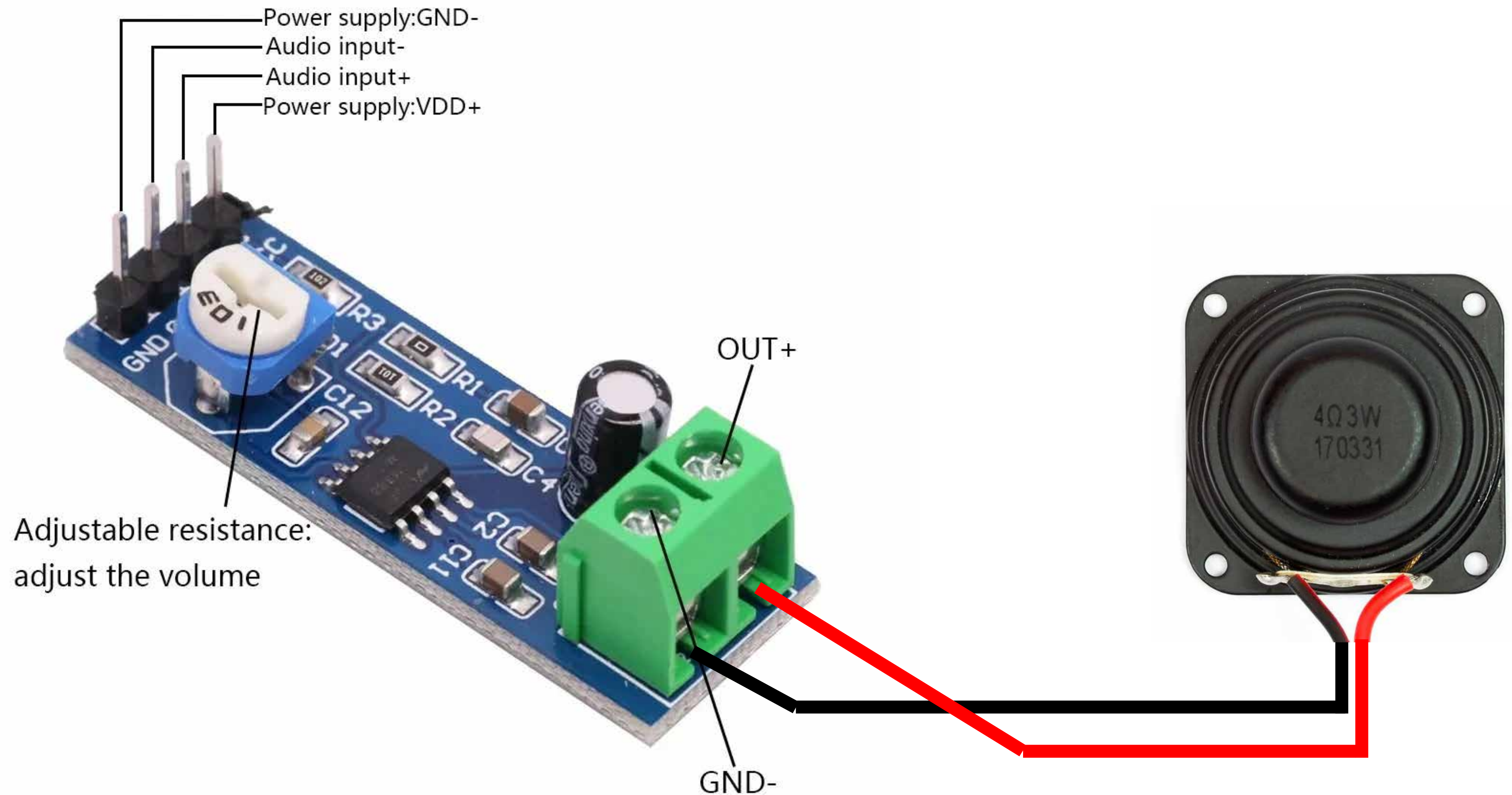


AMPLIFIER MODULE

- on-board LM386 Chip
- on-board speaker wire holder
- on-board 10K variable resistor, you can adjust the amplification volume



connect the speaker to the amplifier



Installing Thonny IDE

- Thonny = IDE for Python and MicroPython
- IDE = Integrated Development Environment
 - application that helps coding efficiently

 **Python** = interpreted programming language

- uses significant indentation
- open source, packages <https://pypi.org>

- **MicroPython** = lightweight version of Python
 - optimized for microcontrollers

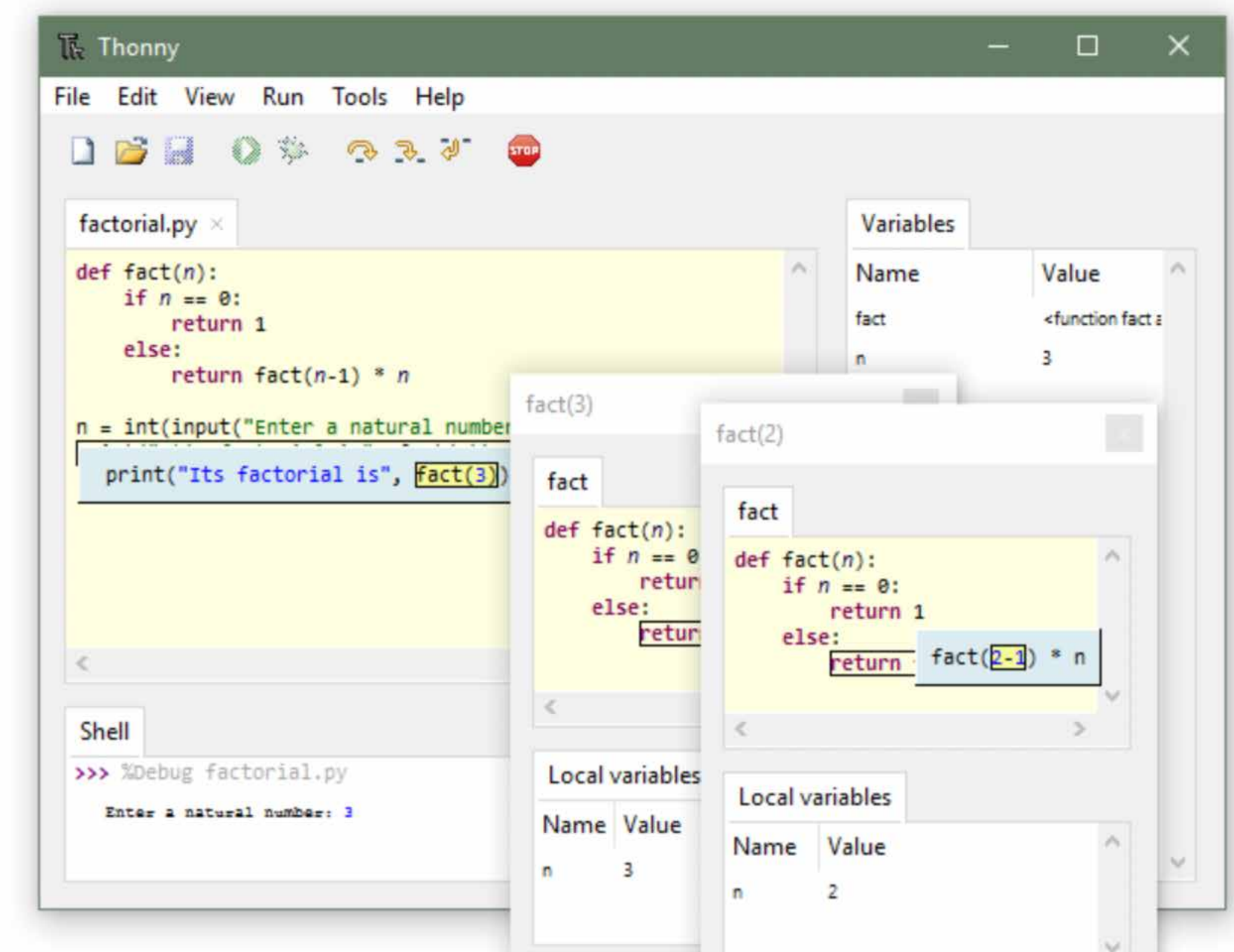
thonny.org

Thonny

Python IDE for beginners



Download version [4.1.4](#) for
[Windows](#) • [Mac](#) • [Linux](#)



Installing MicroPython on ESP32

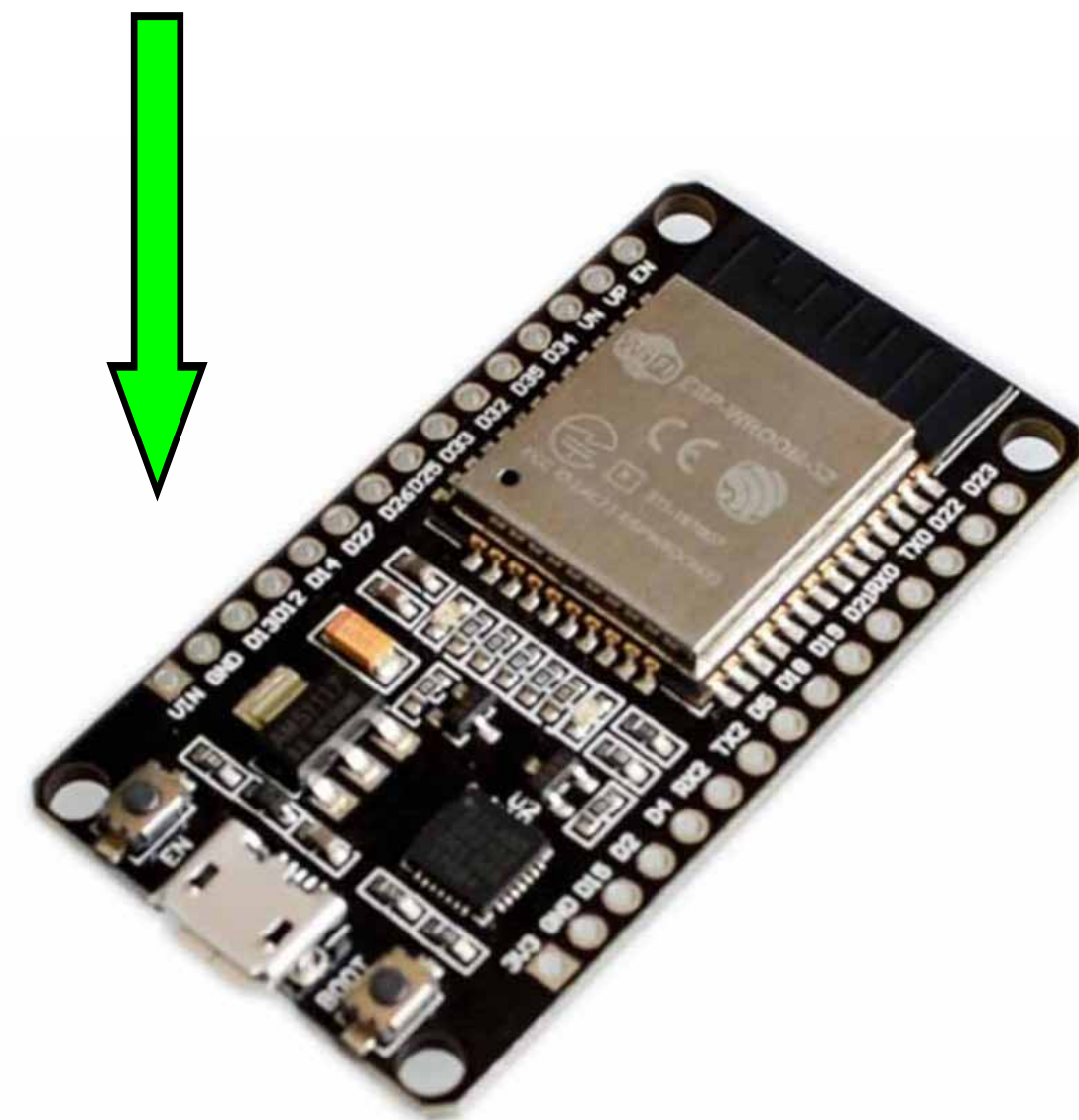
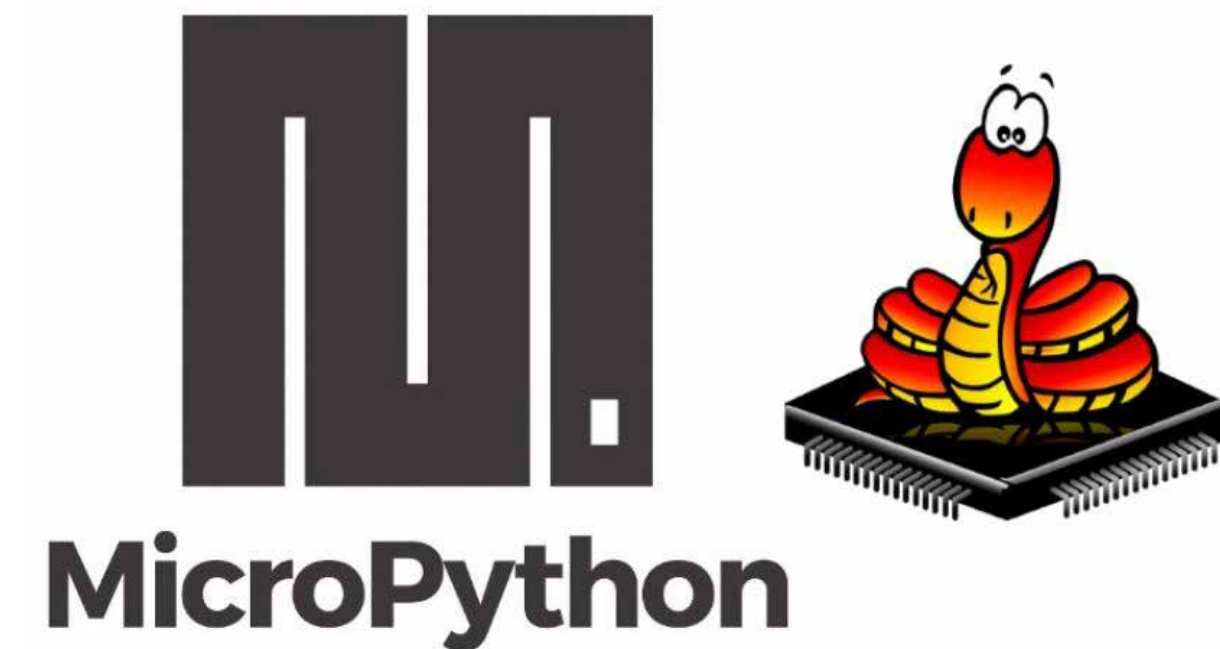
!!!

**WHEN INSTALLING THE FIRMWARE:
DISCONNECT ALL THE WIRES FROM THE
ESP32 PINS**

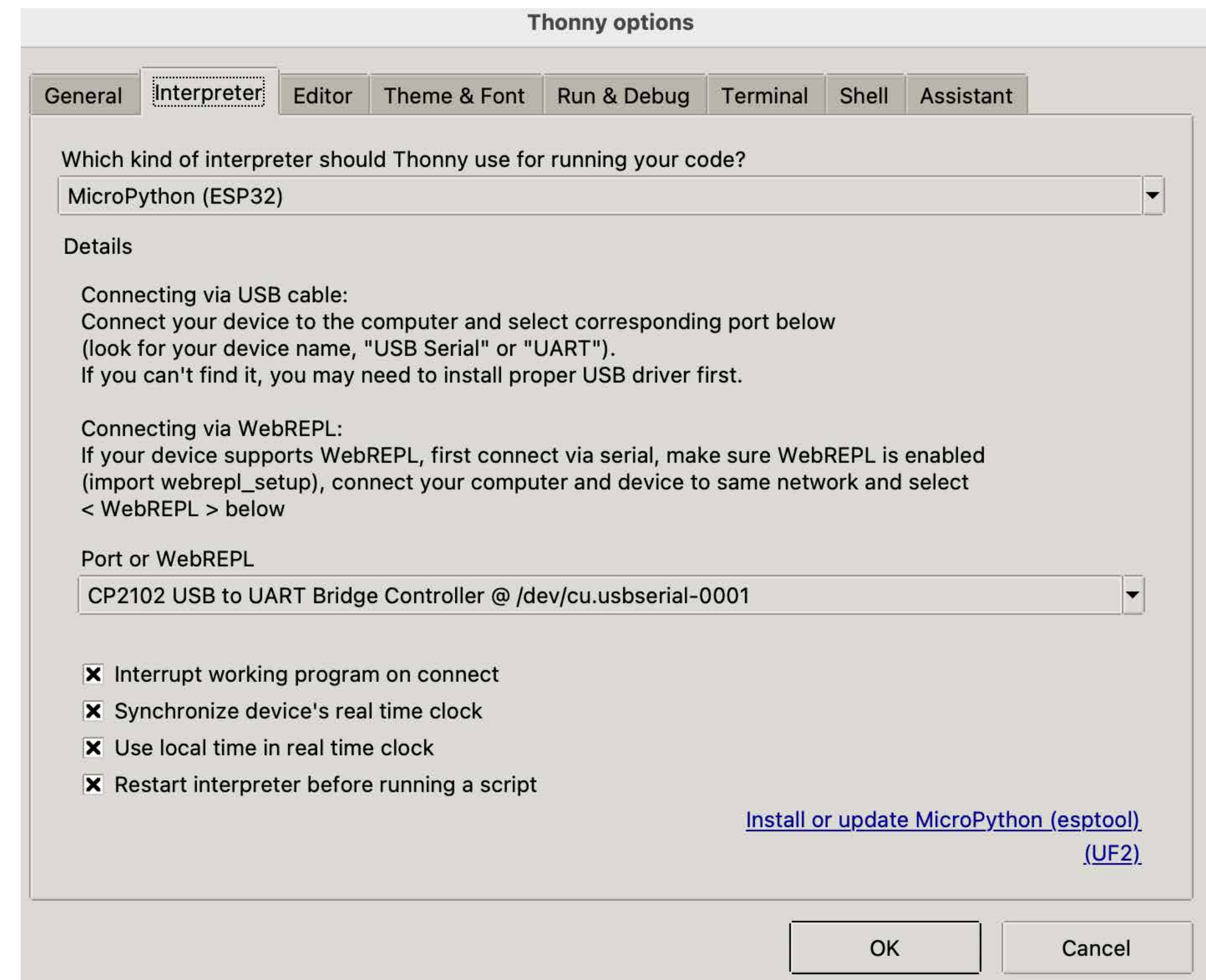
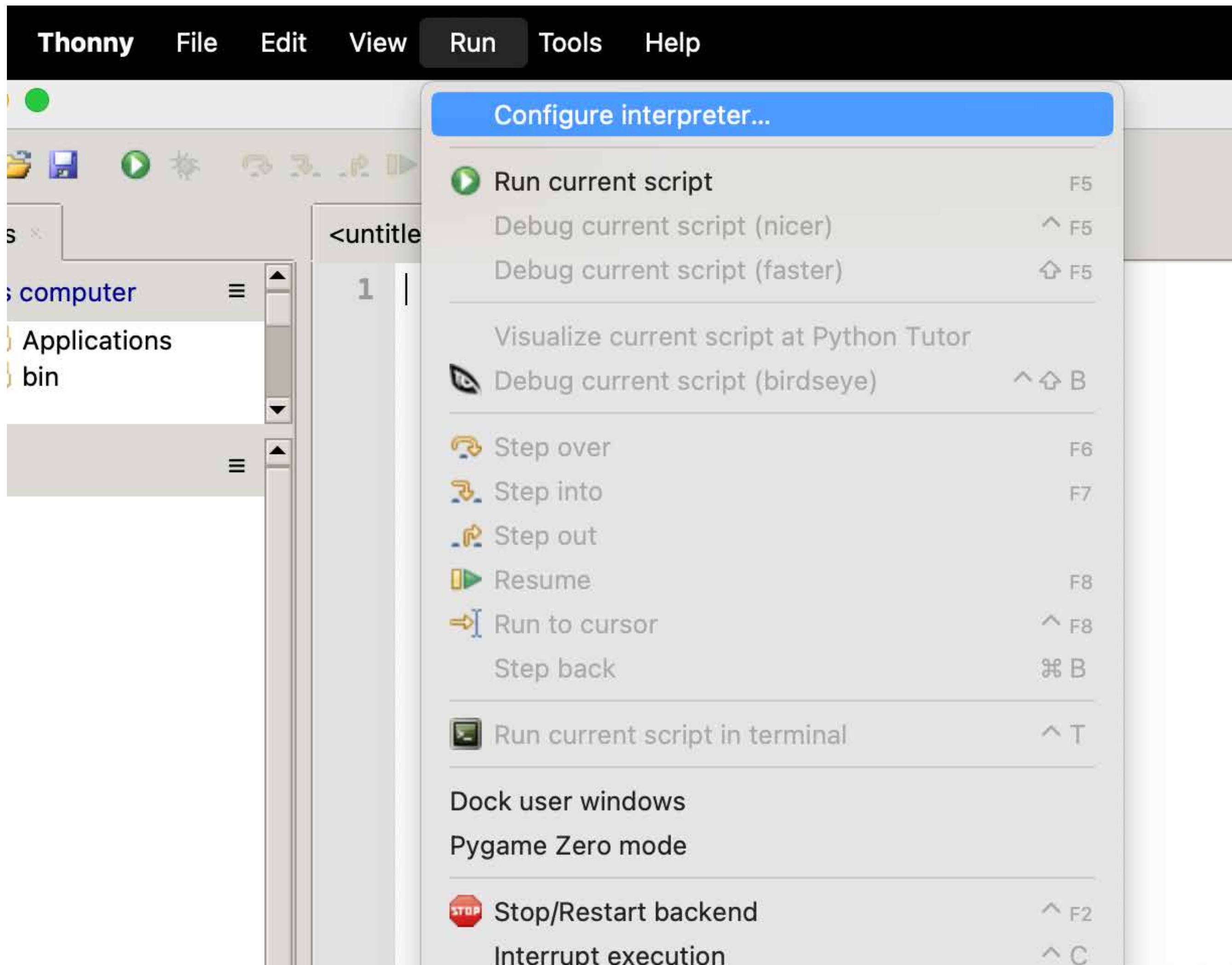
!!!

**WHEN CONNECTING WIRES / TOUCHING
PINS: ALWAYS DISCONNECT ESP32 FROM
POWER SOURCE**

!!!



Installing MicroPython on ESP32



Installing MicroPython on ESP32

Install MicroPython (esptool)

Click the ≡ button to see all features and options. If you're stuck then check the variant's 'info' page for details or ask in MicroPython forum.

NB! Some boards need to be put into a special mode before they can be managed here (e.g. by holding the BOOT button while plugging in). Some require hard reset after installing.

You may need to tweak the install options (≡) if the selected MicroPython variant doesn't match your device precisely. For example, you may need to set flash-mode to 'dio' or flash-size to 'detect'.

Target port

☒ Erase all flash before installing (not just the write areas)

MicroPython family

variant

version

info https://micropython.org/download/ESP32_GENERIC

Target address

Install speed

Flash mode

Flash size

☐ Disable stub loader (--no-stub, some boards require this)

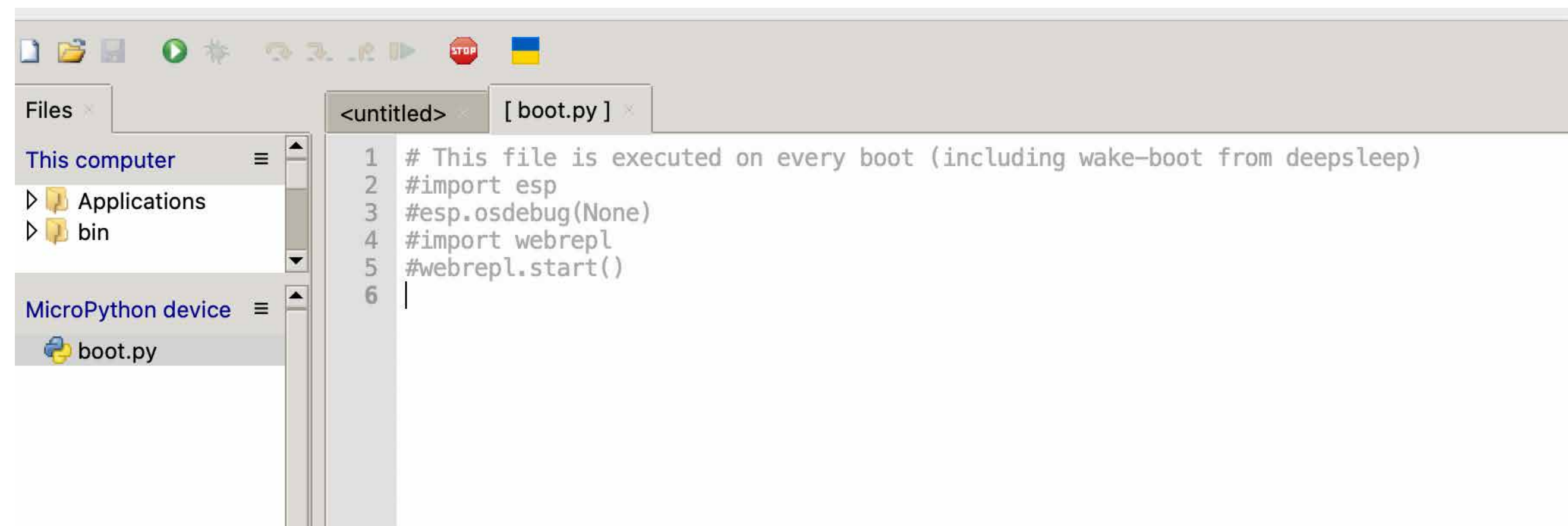


Install

Cancel

Shell

```
MicroPython v1.26.1 on 2025-09-11; Generic ESP32 module with ESP32
Type "help()" for more information.
>>> |
```

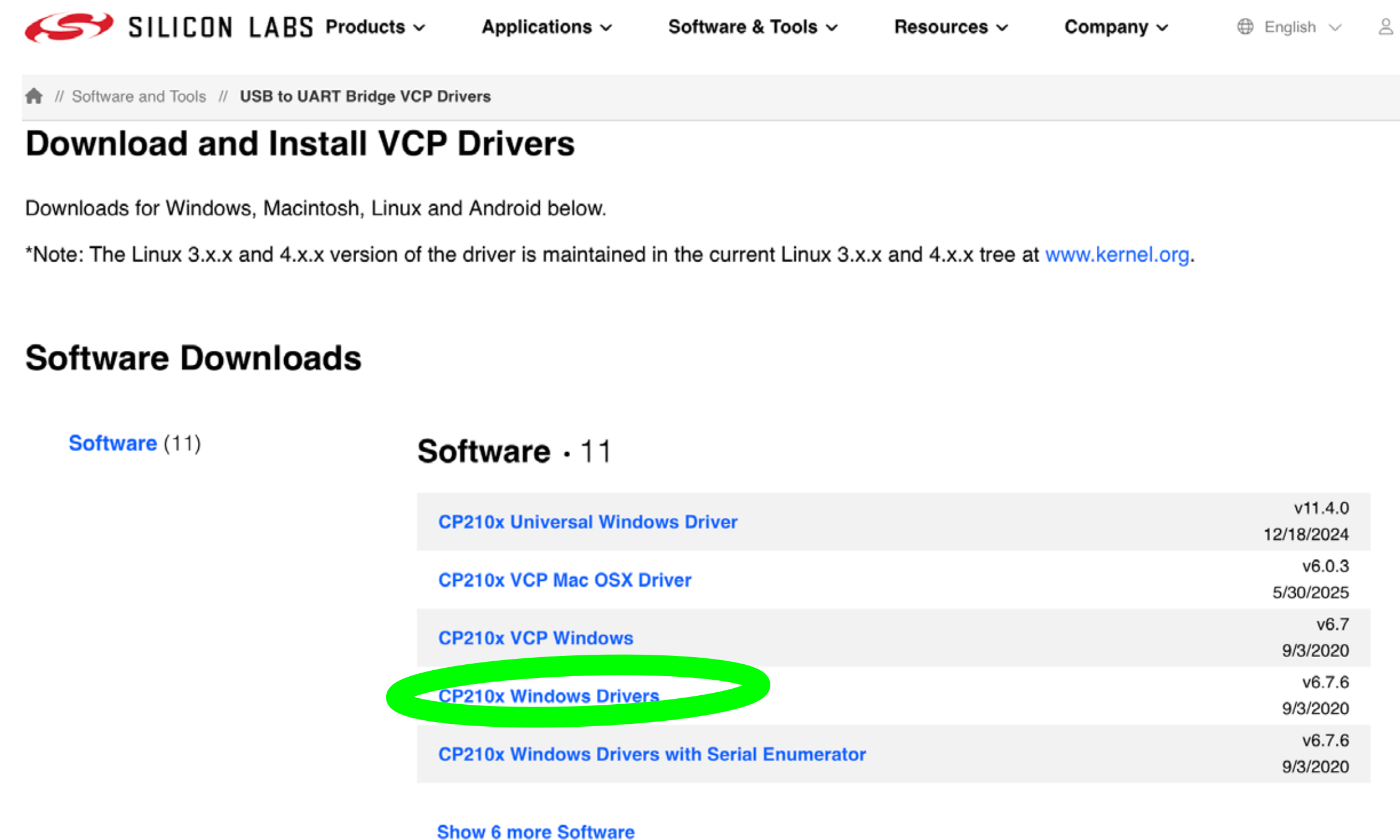


Installing MicroPython on ESP32

troubleshooting

If you get an error message while installing MicroPython on ESP32 (can occur mostly on Windows 11 but also on other versions), please download and install this driver:

<https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads>



SILICON LABS Products ▾ Applications ▾ Software & Tools ▾ Resources ▾ Company ▾ English ▾

Home // Software and Tools // USB to UART Bridge VCP Drivers

Download and Install VCP Drivers

Downloads for Windows, Macintosh, Linux and Android below.

*Note: The Linux 3.x.x and 4.x.x version of the driver is maintained in the current Linux 3.x.x and 4.x.x tree at www.kernel.org.

Software Downloads

Software (11)

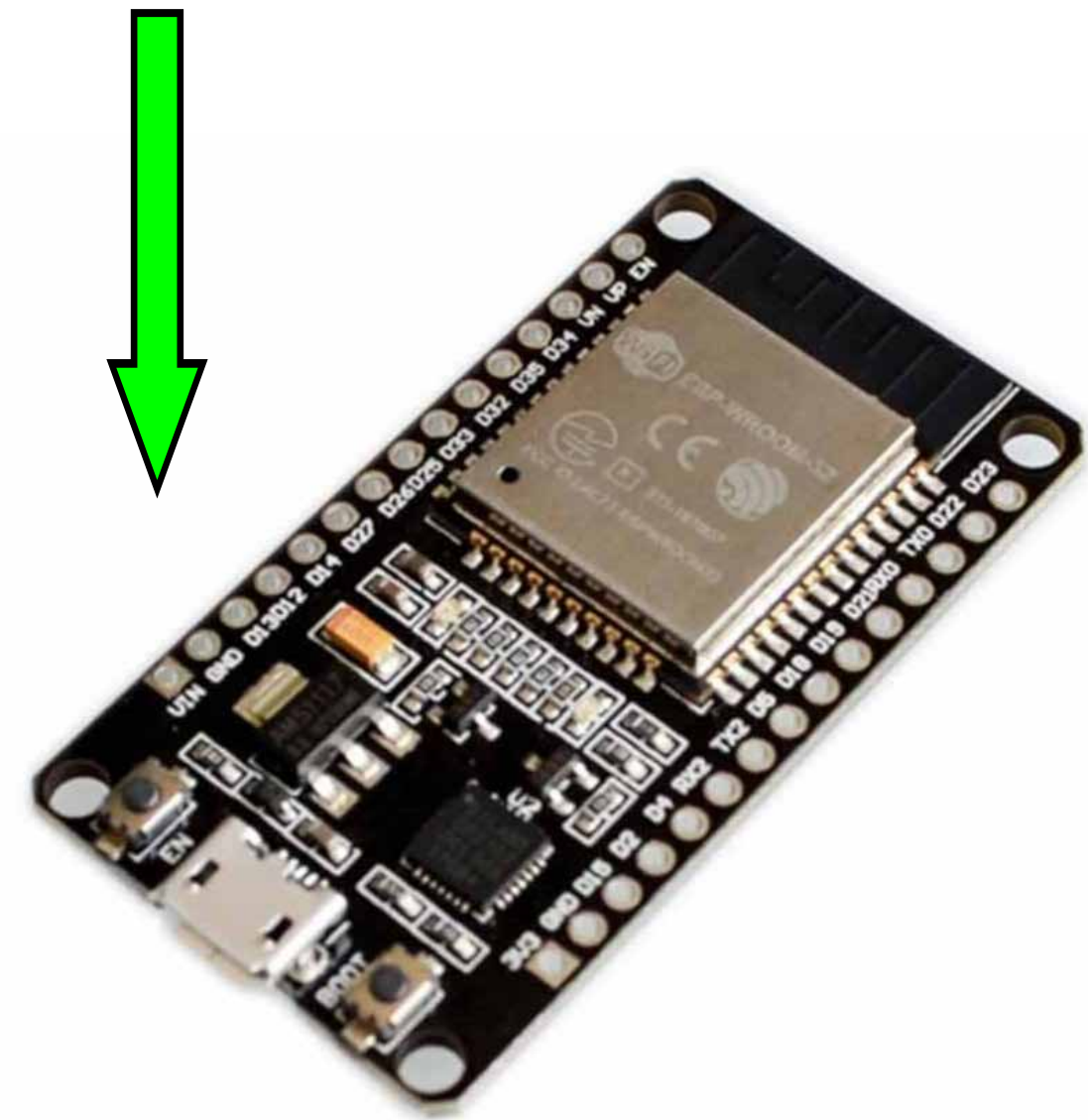
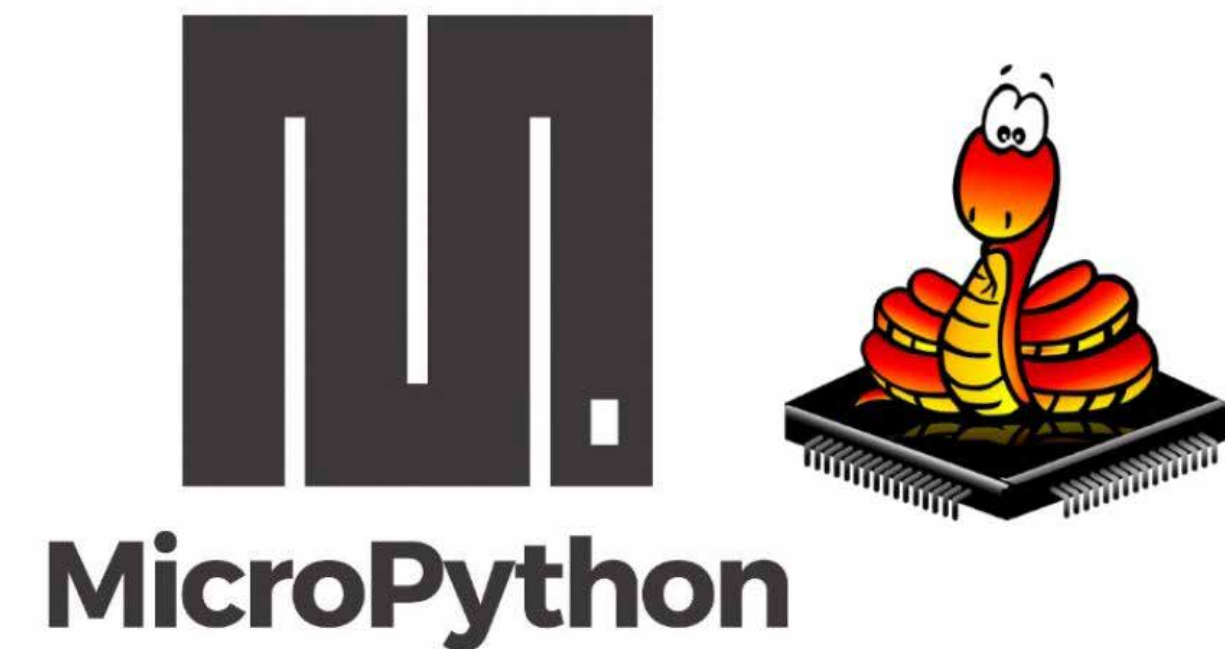
Software	Version	Release Date
CP210x Universal Windows Driver	v11.4.0	12/18/2024
CP210x VCP Mac OSX Driver	v6.0.3	5/30/2025
CP210x VCP Windows	v6.7	9/3/2020
CP210x Windows Drivers	v6.7.6	9/3/2020
CP210x Windows Drivers with Serial Enumerator	v6.7.6	9/3/2020

[Show 6 more Software](#)

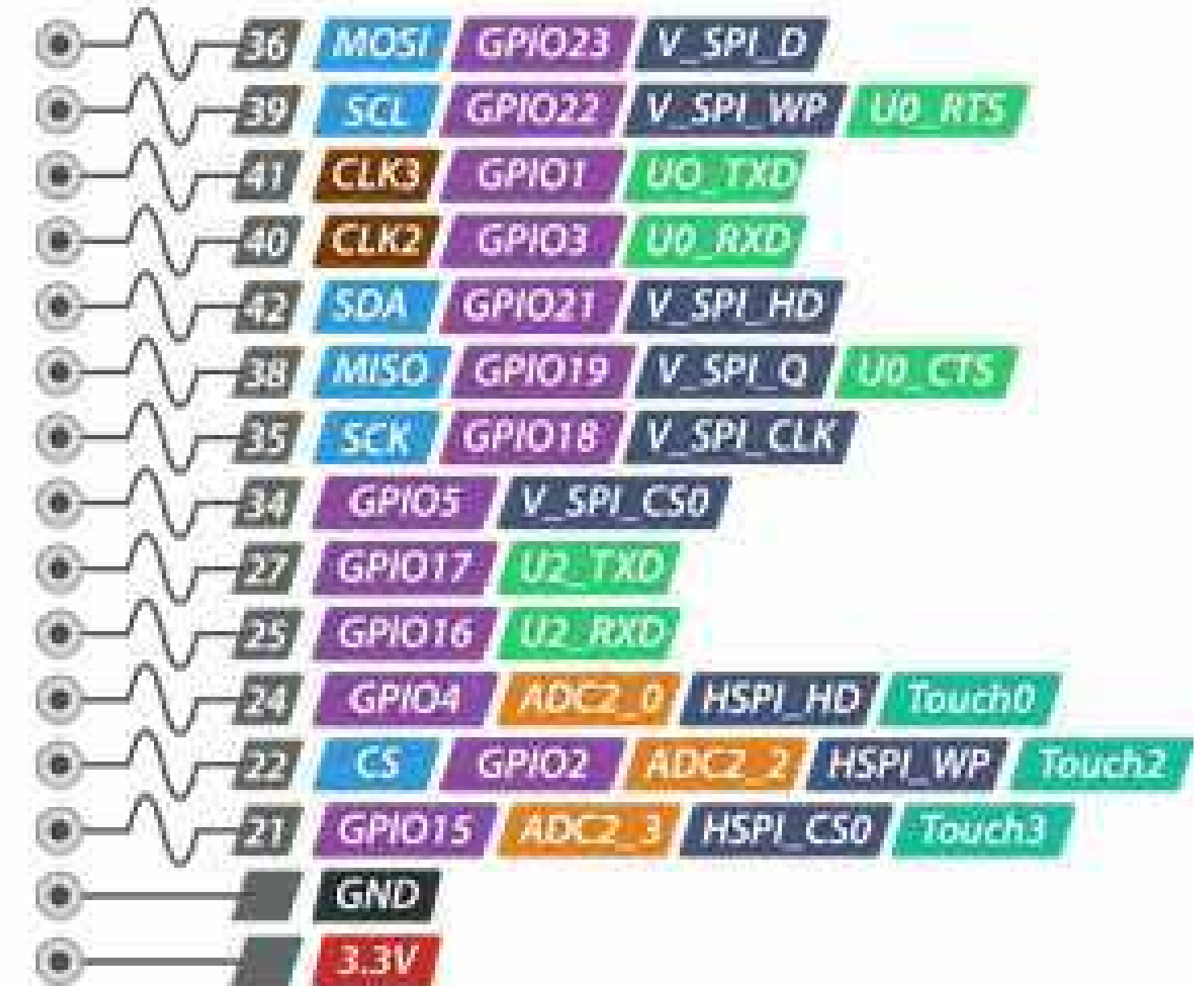
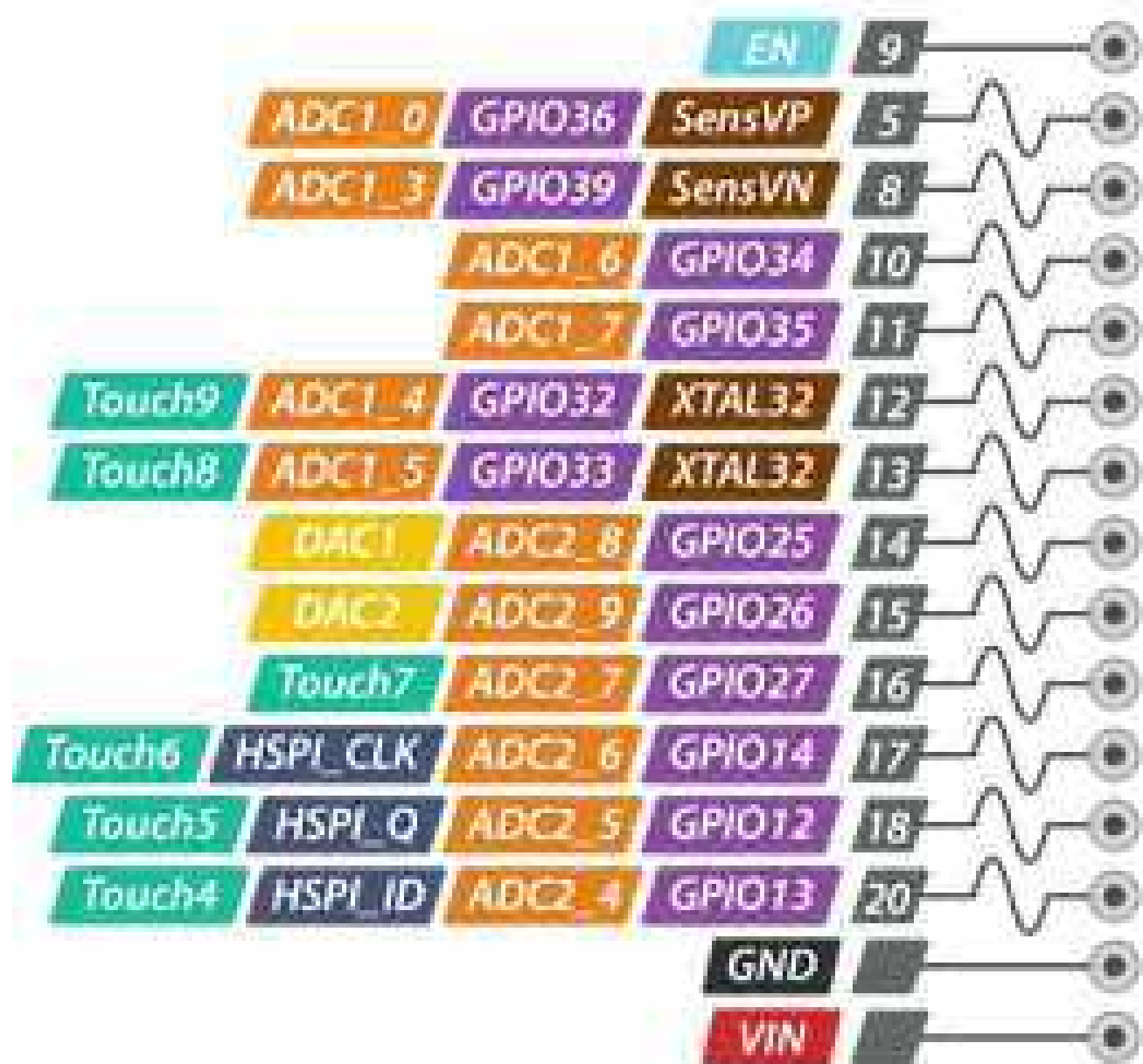
Installing MicroPython on ESP32 - option B

<https://micropython.org>

- 1) go to “Downloads”
- 2) select the ESP32 WROOM board from the list
- 3) download MicroPython as a .zip file

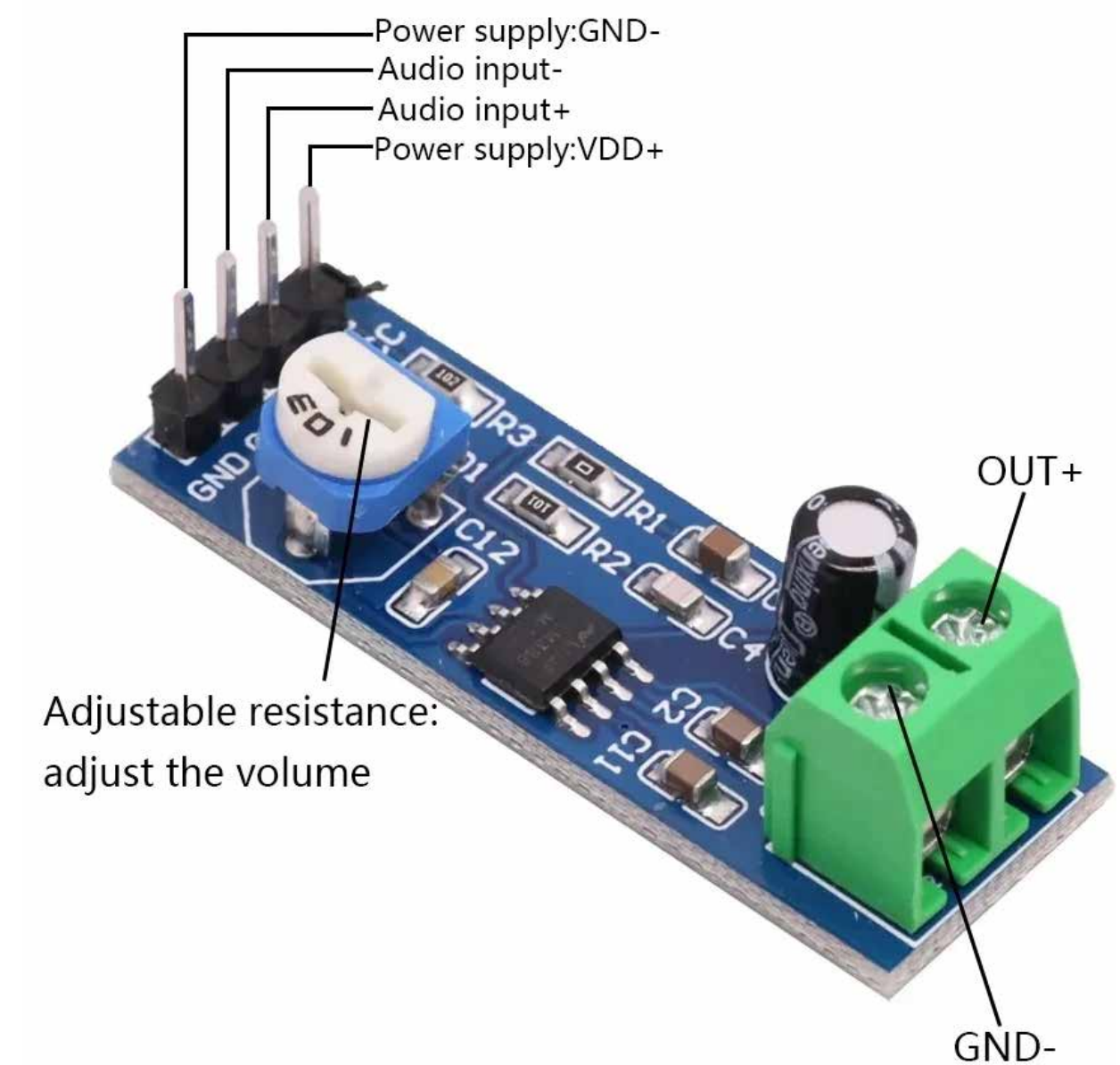


ESP32 PIN OUT



Wiring diagram for Creative Hardware Workshop speaker + amp

2025 by Silvia Binda Heiserova

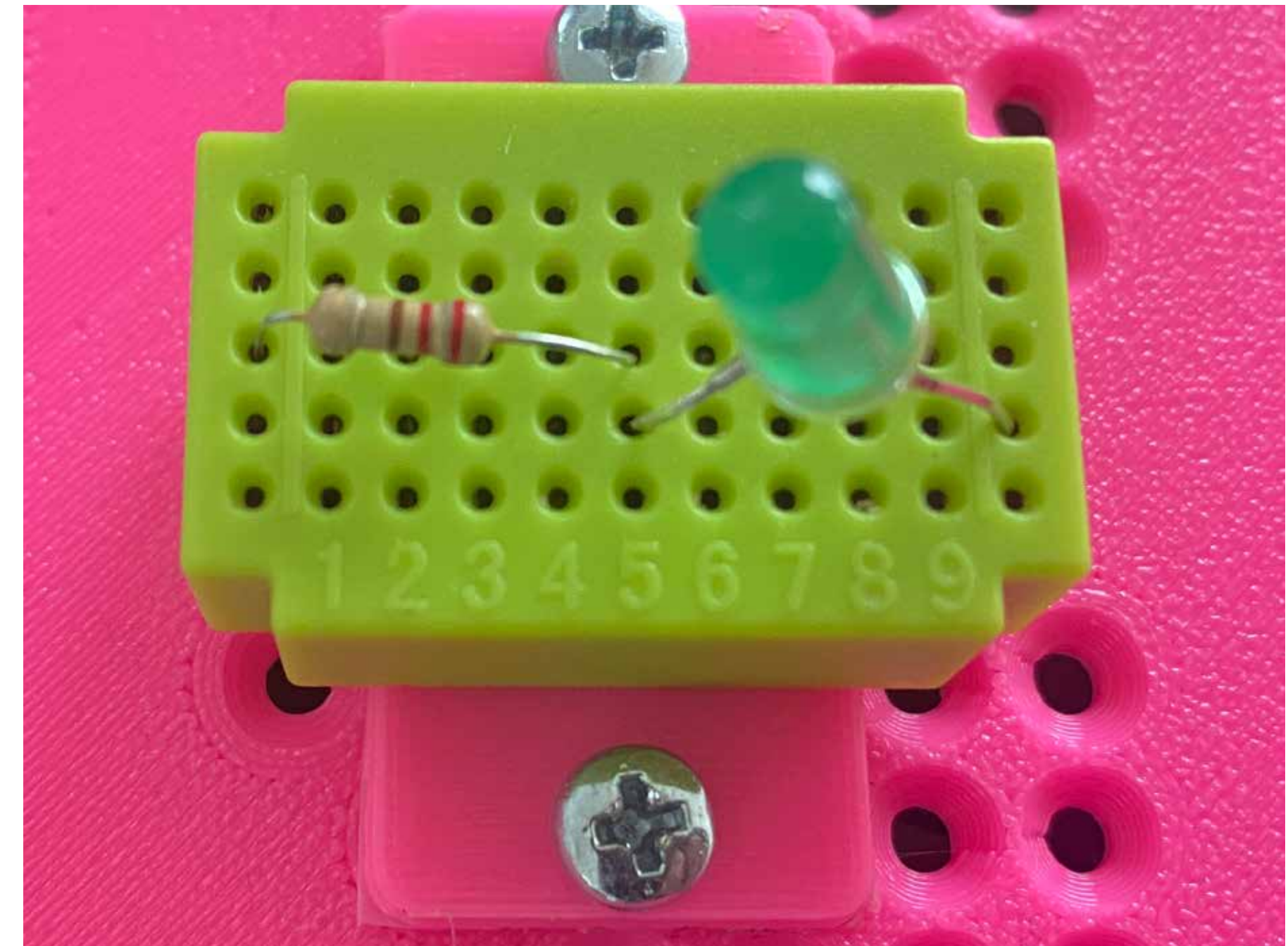
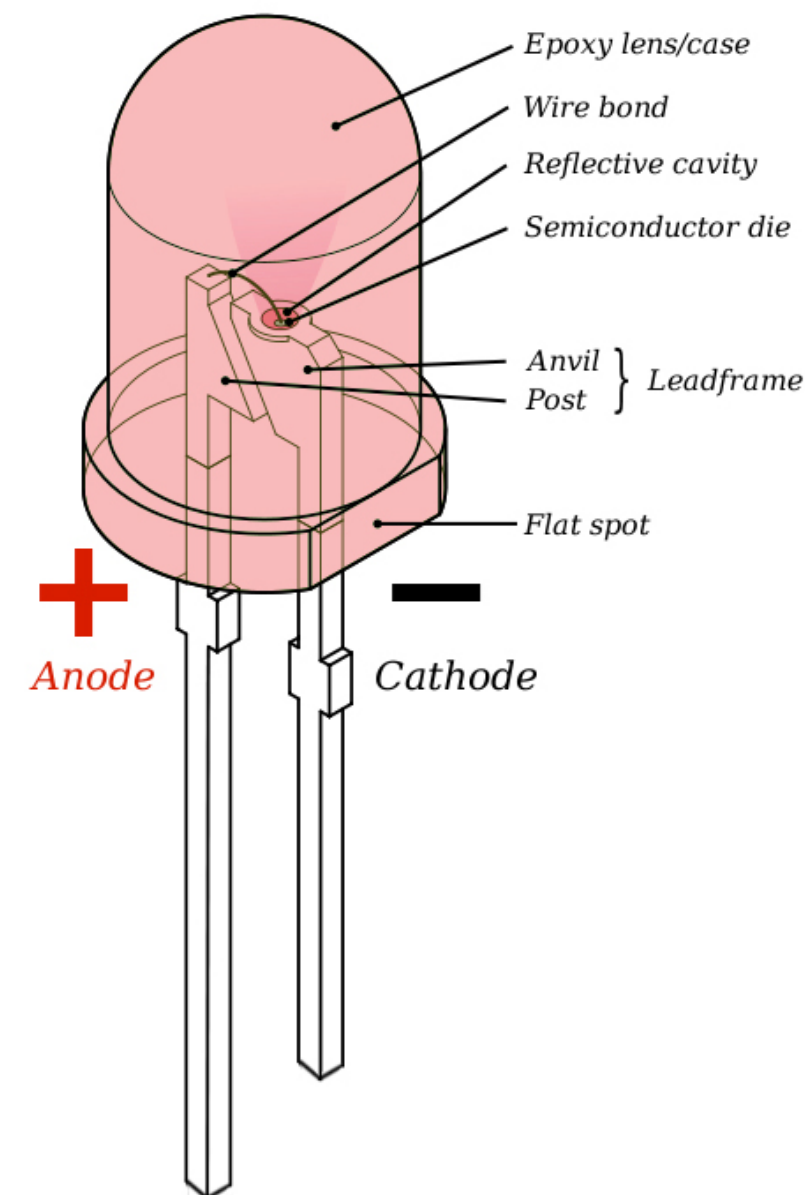


Wiring the ESP32 and modules

1) **disconnect the ESP32 from power source (USB cable)**

2) **assemble the LED MODULE:**

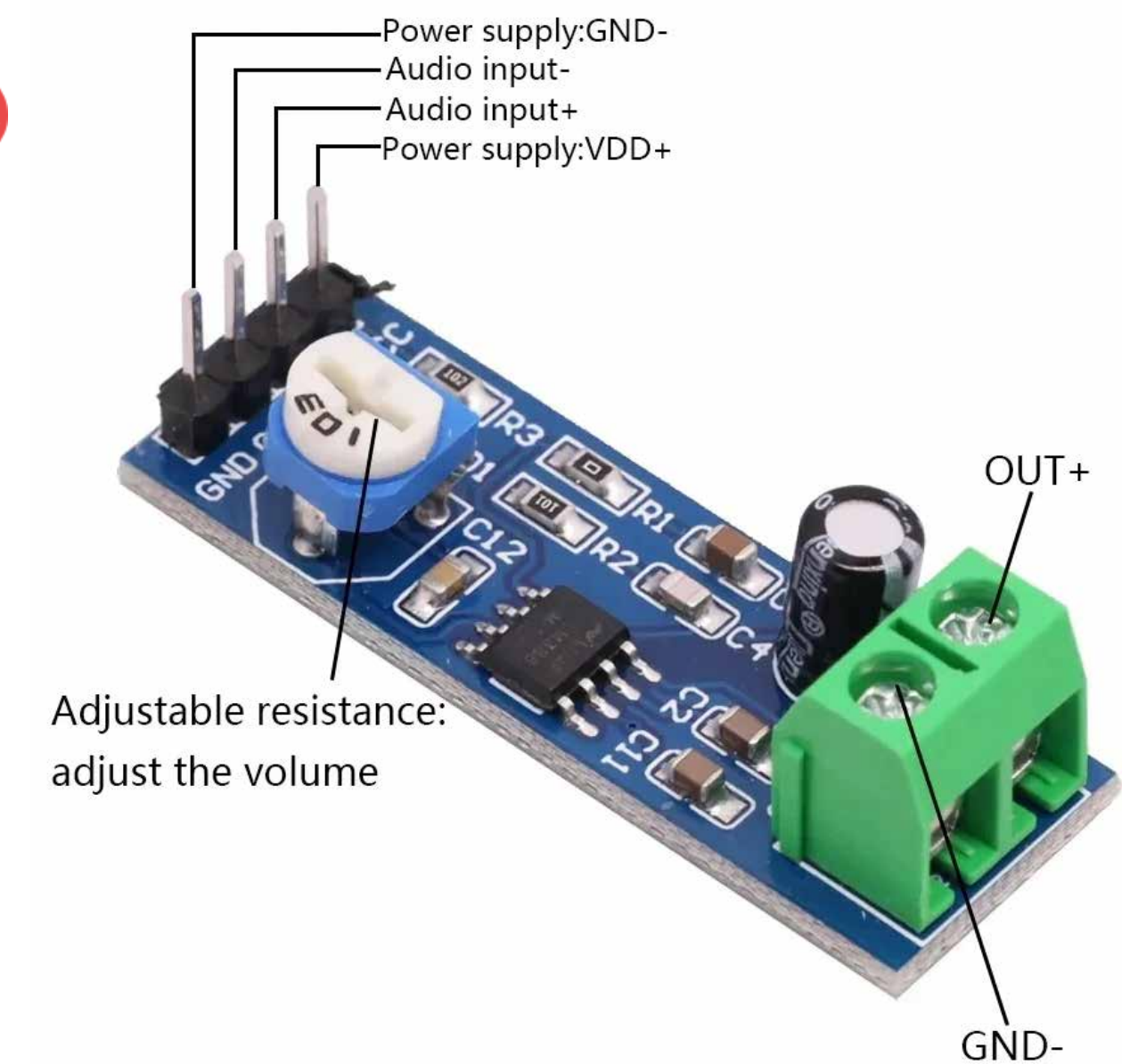
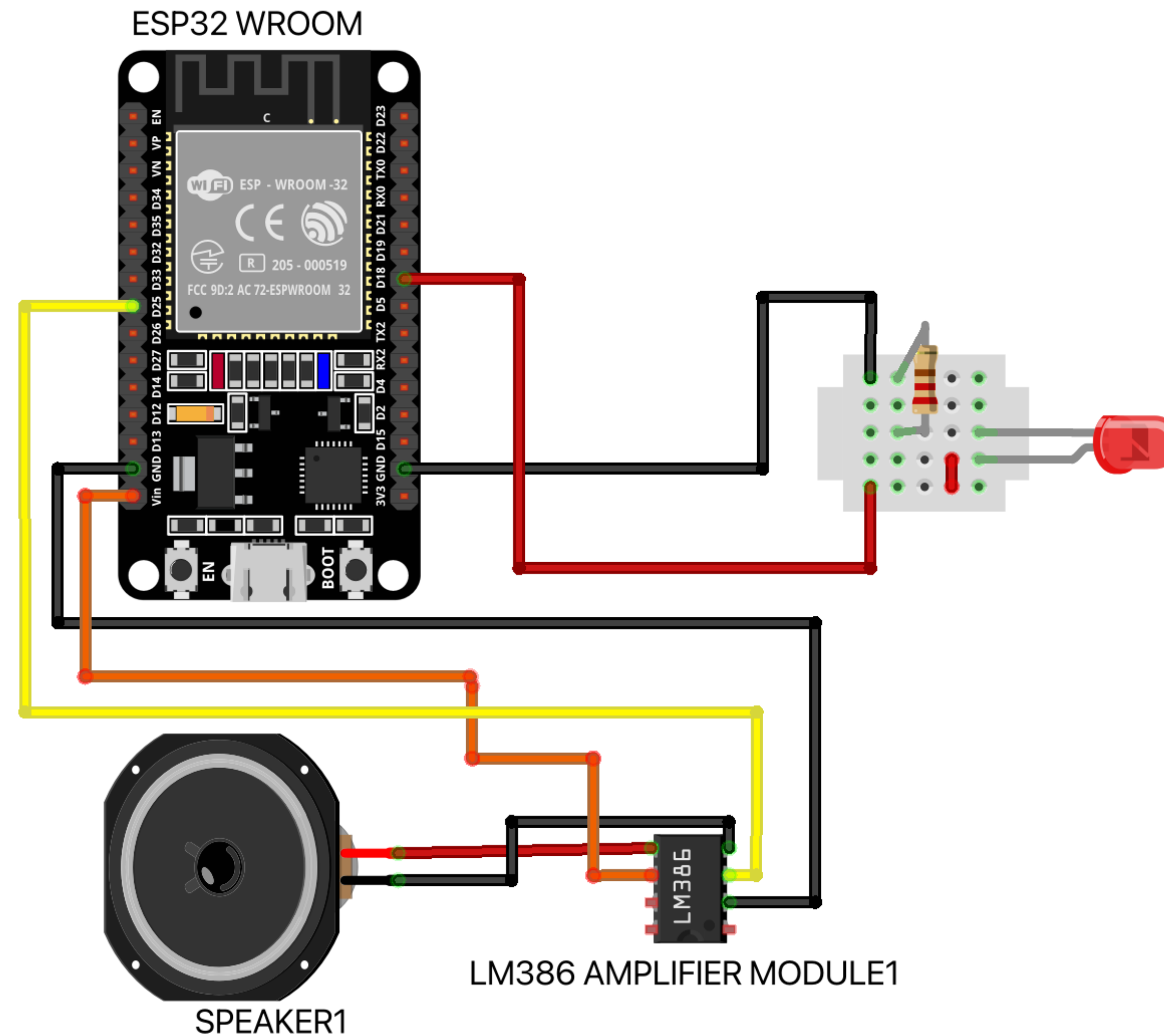
1x breadboard
1x LED
1x Resistance



WIRING SCHEMATICS - connect the LED

Wiring diagram for
Creative Hardware Workshop
speaker + amp + LED

2025 by Silvia Binda Heiserova



Uploading Libraries to ESP32

LIBRARIES

- = collections of pre-written code and functions
- extend the capabilities of Python/MicroPython
- provide a wide range of tools and modules
- making it easier to work on specific tasks without reinventing the wheel

FOR EXAMPLE: “i2c_lcd.py”

- = library for running an i2c LCD display
- copy it and save to ESP32



RESOURCES:

- <https://www.makerspaces.com/basic-electronics/>
- <https://opencourse.com>
- <https://www.instructables.com/Basic-Electronics/>
- <https://www.allaboutcircuits.com/technical-articles/an-introduction-to-ground/>
- <https://www.circuitbasics.com/introduction-to-microcontrolleres/>

CONTACT:

silviaheiserova@gmail.com